

TSIE
IDIVT
PRRRK
AGITLL
EFKRIV
RPQQVP
STIHQK
LKEAIRD
LKLTTAD
SHLMSSL
VLRRHLG
LIWVIWRR
RIWVIWRS
WIPAQGEN
ARHWKMLWY
AVNTEKALA
DINSGVESA
DRIAKLIPA
GRNGWYRIN
IEIPSSVQA
KKWNRRRVK
MGFTRKWQF
NLLFPINPG
PPMKEEVIE
QISITYDGS
RFKWWRRYW
RIKLLTIWK
RIRYIQRVW
RNVRRNNTR
RWLVLRKRW
SLRRYIWRP
TIKRYIWK
VESAKVKLK
WKKHLLKIM
ERLLKVLGRP
PLPFDVGKKL
EYLSRKGLLEL
FIQGYSDLFGN
IKQLLHFFQRF
INPATGHPWGA
KGYFYFLFKFK
LDEIDDVLEEN
PDTPAAYFSNH
RCNADEQAANA
SLLAYHTGQPL
ILRWPWWPWARK
KKCPVIVIRVCK
LATLTGAIIVAL
LRPAVIVRTKAL
LVHKLILRHLLF
NRWRIVVIRVRR
RLAFIVVIRVAR
RLARIVLIRVAR
RLARIVVIRCAR
RLARIVVIRVAL
RLARIVVIRVAR
RRAAVVLIVCRR
RRWRIVKIRVRR
RRWRIVVIRVKR
RRWRIVVIRVRS
RRWRKVVIRVRR
RRWRYVVIRVRR
RWWKIWVIWWR
RWWKIWVIWWR
RWWKRWVIRWWR
VPPVKNPYQPNF
VRLGIRVAVIRA
VRLHIRVAVIRA
VRLRDRVAVIRA

VRLRIRVAKIRA
VRLRIRVAVIKA
VRLRIRVAVIRP
VRLRIRVAVPRA
VRLRWRVAVIRA
VSLRIRVAVIRA
YDPSIIAKFAIQ
YTSVITIELSNI
FLKLLKKLAAKLF
IKPWKWPWWPWRR
ILPWKWPWWPWSR
ILPWKWPWWWRRR
ILPWVWPWWPWRR
IWPWKWPWWPWRR
KNKGKKNGWWWWW
KVAKNVESTFRF
WRWWKPKWRWPKW
ALYKELDLEGAFME
DFKAQNVTLAILQ
DMHNGYKWSNTMFS
HDLETFKKLIGRDC
IFNAIFWLTQILFN
IGSLESEDFADYYC
KLKKALRALARHWK
KLYRKFKNKLLKLK
MEVLEELNALANAN
PAEFVAKEKEIELA
PGYPRDVMPPPDLP
RPLWSHLASSRRNT
SAGAELLSQDSDNK
YAARIEALLRAAQE
YSQQFLTTFQQWDI
ADPRNPLEECFRETD
APPVLSGENYKASCI
ARDLMRKPHIIIATP
CTVAGWGRVSMRRGT
DEIDSVLEENAEFEV
ELLGEEVKVSLPGGD
ELQKLNWDVFTNWL
ERQVTYHEGKELAQR
GTKVRQGTLLKARYN
IHKCRSPDKETFTCW
IKTSPPIVVYRESVT
ISPEFPTSGEGLMEN
LIVFLTLLLCGGLGL
SRPIARFYTWSQNTT
TGIFFGMKAPLMQDS
TVEKTVEHLEAEVTG
VGALAVVVWLYLWLW
VSGQVSPLIDWWNG
DEEHRTWFCARYAWYC
ECNYTLEQQAPFIRAQ
GFSQKSHLIRHLRHT
IKTLKSAKRFKITAS
KWQHSIKWQMRLLHLV
LLLLILLSLSNYYSIQ
YIVYKIRFAFKRSKAL
DYVIPGNDDSAKAVALY
FTLEGNPDTGAPTKVLA
FVDLKKIANIINSIFGK
FYDPLVFPSDEFDASIS
GVLDAFKKIATVVKNLV
HPHNPHVPTSHANVRFF
KCLKKLGKLLKLLKLC
KGGLKLLKLLKLLKL
LFRLLKWLKVKQFGE
LLELLESRLDNVVYRMG
NIVETAKRTGARVVGPI
SLHSLAHETRIASFLSS

SSLLEKGLDGAKKAVGG
TITLPPNSSNHMKQWYT
TTTVNVDFVNSLSLEPC
YIVYKIRSAWKRSKALK
ERLENYKNETIHFEEQRA
GLWRKLRRKAKKVLOHIL
KNLRRRIIRKIEHIIKKYG
LGRKIGKTGRKVWKAWRE
MEEKQEYLKQGYNVIMD
RGFDETFAPHSRYADFP
RVAYYAKWIIHKIILTYKV
TEADVQLELDRRKPGTSR
EKGLDGAKKAVGGLGKLGK
FFRLLFHGVHHGGGYLNAA
GGVAYLLKADHSLDLPRRL
GKPRPYSPRPTSHPRPIRV
GSAQPYQLHKVWNWDPYG
GWMSKIASGIGTFLSGMQQ
MLGDVWEWTSSPLRPWPGF
PDSDLCLKFAMLC TLHDKC
PVSALLHGEYGEEDIYIGT
QIILTCDRYPREVEGLEPR
TPKSASPIVLKLLKSAIAN
ARLARLALTD AELDGYAGQL
FFFGKHKNKGKNGKHNGWK
FSQLEAAKKLDEFSDLDEDL
GLFKALLKLLKSLWLLLLKA
GNGKGNPEYYVAEIQPGKML
ILARVGDKQALGAHGEELAA
ITLNNVALDPIDISIELNK
KKMAVEITTNKVQRPVCN
KWTNTTETGAPQLNPIDGPL
LGVGIAFLATVVTQLLVYNG
LPLGLTLIFYILVSNLIGVA
LVKDSGQAVSYPLDMKATY
PERAAEEISQLLYHAQVLMML
PETTTGGNALKFYASVRLDI
TGGPNGSIRFPQEYSHAANA
TKKTSLEKFAVKSCLSQLYQ
TTTTATATAVTTNPTSFEAS
VKIALFDGSYHDVDSNENAF
AFCWNVCVYRNGVRVCHRRCN
GFCFYVCASRNGVRSCYRRCN
GFCTNVCVYRNGVRACYRRCN
GFCWKVCVYRNGVRVCYRRCN
GFCWNVCVDRNGVRVCHRRCN
NMTLTELKVIADIYPGQTQF
QHAFYQLIHQGTKMVPDFIA
RRGEWLRRHVIWTYMKDYFPIT
DLAAEMYQEGRTVLRDAGFRQYE
FAKKFAKKFKKFAKKFAKFAFAF
FEFRDHIVRFLNMFILDSGYTIQ
FLPLVRGAACLIPSVVCAISKRC
FWGTLAKLALKAVPAVMGMIKKE
GKSGCTVIFLNQLRLKIGVTYGN
VIAYEPVWAI GTGLTATPQQAQD
VIEAITPIIGWLLGRAALQYVEA
AATAEISRVQNWQWIRYGVELDGD
IAPARDFAWTRDKAIAFAEENNP
ITLNNVALDPIDISIELNKA KSD
MGIGKFLHSAKKFGKAFVGEIMNS
RLECAAIQKLVAQKRANRRSSKST
IWL TALKFLQKHA AKHLAKHSLSPW
IWL TALKFSGKHA AKHLAKQQLSPW
NQGRHYCGGALIHARFVMTAASCFQ
RLLMGGSQGAHALNLAVVNMLPQL
VDGAEALRSATENRPDAIVLDINMP
AGGDWAGLFDAGADPKPDHVL FSEFG
ILITAGPTREAI DPVRYITNHSSGKM

LRKCCEDGMRENPMRFSCORRTRFIS
EKYD TVDNVEEDTLTFTLRNTWIFRPD
RG CYKICGETCLFIPCLTSVFGCCKN
VIAWASCGTAGFKGSRKGT PYAAQMAA
YTISSILVLTFAALTAYESRMISVISQ
FNGFSRFRFCLGDFTLFLGNNRCWLFIG
GLWKSLLKNVGVAAGKAALNAVTDMMVQ
SSLLEKGLDGAKKAVGGLGKLGKDAVED
GEYCGESCYLIPCFTPGCYCVSRQCVNKN
GRNDDDGYYFKFGIYRVGNSTVPVTYNLS
IIMPLYQLVLLFVLFVGF AAILQVPDLNG
SAQITAAVALVEAKQARSDIEKLKEAIRD
AQIVEASKEDARKEGEK LIEQARAEIQQER
GIPCGESCVWIPCLTSAIGCSCKSKVCYRN
IEAQQHLLQLTVWGIKQLQARILAVEYLK
KHVRDCPKGIWRS CRYKCIDNKC VFTY WPH
MDKGEVADLTAWLEKVK EIQERYPLPEEPK
YVCVAPLSNIQGLTWQNSLVSHKLQRVLWV
GLFLDTLKGAAKDVAGKLEGLKCKITGCKLP
IAQAIQKIKEPEFEYLKDYIVISRVLISKDR
SALRTGFGSAIYFTSLNALRQNVARSNLLRT
TIVGTVAVENPSLVQELCQAFPSKIAVGIDA
LVQRGRFGRFLRKIRRRFRPKVTITIQGSARFG
QLAPGTEGGFTTNFLNGEVTSIFDASTAYKEA
VSCIRNGGICQYRCIGLRHKIGTCGSPFKCCK
ALWKTMLKKLGTMALHAGKAALGAAADTISQTQ
AMHAVIDRQKNHGMHFRVLAKALRLSGGDHHA
EVNKIALLLSTNKAVVSLSNGVSVLTSKVLDLKN
GGTIFDCGESCF LGTCYTKGCSCGEWKLCYGTN
GIFDVVKGV LKGVGKNVAGSLLEQLKCKLSGGC
MRFL LACFLIVANHILWVRFFQVHEFPKPRGL
GLFRRLRDSIRRGQOKILEKARRIGERIKDIFRG
GLVRKGGEKFGEKLRKIGQKIKEFFQKLAL EIEQ
IHRQLVMGRKIRRL EAISNSIKTSFHIAKLKAEL
LPIVISEAEGVWKDPEG NKYMDMLSAYS AVNQG
LQEF LFGDLDEDTRRRAGINPLRVLDDKRPELR
LVVVEDFGKYKVLGSTRDDAAGEAFDKAAKMLGL
NKCNGTDAKVKLKIQELDKYKNAVTELQLLMQST
PNFFGLLEEEVREIGEIAHEAGAYFVVGVDP TIL
APAVCTAASSTIDDQSSFSNWGSIVDIYAPGSSI
AVALVEAKQARSDIEKLKEAIRD TNKAVQSVQSSI
ESPAEVL TLDILLRRGKTVRPKTLNQKRYVDAIDA
HFHVSRCMTDDDDCAPEKY NIRC RKGF CVQIRKY
NANEPLFANPRNAASGSLRQLDSEITKKRKLQFIP
NCPTEDGGWQVHGVT SFVSAFGCNFIWKPTVFTRV
SRPSPQVP IRPSQPRPQPFKPRPQQVPPRTPHPRL
YIPLHVSSEQLGA AVAGLKT LGASGWNVTIPHKTA
ILEC IFDIDCPTKKCAPPLVAKCDMYECYCRCPNN
KVLMLNDDYTPMEFVVHVLERFFAKTRDEATSIMLQ
KYYGNGVTCGLHDCRVDRGKATCGIINNGGMWGDIG
LDDIPGTSVTNFFRSIRSTLDFDFEDENEGTWEPEKE
LKLK LKLK LKLK LKLK LKLK LKLK LKLK LKLK LKLK
LLDNFESTWEQSKELWEQQEISIQNLHKSALQEYWN
MKKLLFAIPLVVPFYSHSDPKGLKLLKGLKLLKL
NHTFEVENSTLNGMDLIERQIKILYAMILQTHADVQ
TVMIGGEPYTLGLFD TAGQEDYDRLRPLSYPTDVF
YHLCKTRFDCPRTYLLFFPRMWKINRRCRYVYFFE
YTSLIQSLIEESQNQQEKNEQQLLELDKWASLWNWF
GFSSIFRGVAKFASKGLGKDLARLG VNLVACKISKQC
GKFSGFAKILKSI AKFFKGVGKVRKQFKEASDL DKNQ
QIFEVDGDSHKLVD SLHGHEGPVWQVDWAHPKFGVIL
SDDECKIDGDCPISWQKFHTYK CINQKCKWVLR FHEY
VSENLALKNVELLKGDSYGCTINIKVNQQKKLDFIII
WNPFKELERAGQVRDAIISAGPAVATVGQAAA IARG
HGPNGEDGT VQGLLELMNIPYVGNGLASAAGMDKVVM
IFLFRAIWLALSLLILFFSMHRLSLLDSTRDVSELISL
NHVTCRINRGFCVPIRCPRGRTRQIGTCFGPRIKCCRSW
TTPVCAVAATAAAASSAACGWGGGIFTGVTVVVS LKHC
TTWEAWDRAIAEYAARIEALIRAAQEQQEKNEAALREL

TYRSSGAGGQHVNTTDSAVRITHIPTGIVVECDERSQ
YDLLRPSSVHKHGSDESSESHEKSSTCDESSWSEACETHE
ATCDLASIWNWNHTLCAAHCIARRYRGGYCNSKAVCVCR
ATGWGLEISDLKKQLEEARSKGISVRALVVINPGNPTGQ
EEAKAAAAKAEAEAKAKADAEAKQKADAEAKAKAEKAAK
IFECVFSCDIKKEGKPKPKGKCKTGGWRCKIKLCLKI
KVSVKPPLNVATEKESTPPKKIEYKPEISNWINLIGFVE
NGYRNIKYCFIDTDCPRSMCHYPEIVRCVDQCKCVRIMP
AVGAWLLCLLLLGLALQGAASRAHQHSMEIRTPDINPAWY
FEKNFGTPQEFNIAAAATAFSIEYQCPPSNAAFSESMAA
GLGGLHGNKGAIIVRFLVDDTSYCIVNCHLAAGQSNKAAR
QITKLPCVTVDDBPKVEKPIPMVAKCFGRFSRHCHYFYF
DDVKIKCVSAIDCMDLFNLLPIVYKINNIVVEQSSQRLI
ERVVRNPQSCRWNMGVCIPFLCRVGMQRIGTCFGRVPCCRR
EVIVVFKSFPTRVGAGPFPTPEMPEEADRLGLVEYGTVTGR
FYKHLSDHYNDSLIWAHNKVEGKQEYTSLLYVPAKAPWDMF
GVGNPVSCVRNKGICVPIRCPSMKQIGTCVGRAVKCCRKK
IKKEIEAIKKEQEAIKKEIEAQHLLQLTVWGIKQLQARIL
IKKEIEAIKKEQEAIKKKIEAIEKLLQLTAWGIKQLQARIL
IKKEIEAIKKEQEAIKKKIEAIEKLLQLTVAGIKQLQARIL
LDGAKKAVGGLGKLGKDAVEDLESVGKGAVHDVKDVLDSVL
NNEAQCEQAGGICSKDHCFLHTRAFGHCQRGVPCCRTVYD
QQQNNLLRAIEAQHLLQLTVWGIKQLQARILAVERYLKDQ
VLRSLSDIRFTNLSGDVHGYKLEFEFDSNDYFTNKILTKTY
YENPYGCPTDEGKCFDRCNDSEFEGGYCGGSYRATCVCYRT
DRIPCVTNGDCPVMRLPLYMRCITYSCELFDDGPNLCAVERI
GLDGAKKAVGGLGKLGKDAVEDLESVGKGAVHDVKDVLDSVL
IHDQVYFNNNSPPCVTDKDCPRPQFRKSNVRCRNGHCVN LGN
ITRLDVPSLATADLTVKARARNDSDAAITATVSGSVGATSFR
KGIIEALFNKLEVSVDYVATKDLASMHPGRTA AIVLDGQTIG
KGLDGAKKAVGGLGKLGKDAVEDLESVGKGAVHDVKDVLDSV
MLKNMDTLMMDLIENGKDANEVLKMPFHYVLSIYQNKNNDIS
TDQDTAKCVQKKNVCYFECPWLSISVSTCYKGAKCCQKRY
ATCDLASFSSQWVTPNDSLCAAHCIARRYRGGYCNGKRVCVCR
ATCDLFSFESKWFTPNHAACAHCILLGNRGGHCVGTVCHCRK
ATCDLFSFRSKWVTPNHAGCAAHCIFLGNRGGRCVGTVCHCRK
ATYYGNGLYCNKQKQWVDWKNASREIGKIIVNGNVQHGPWAPR
EKGLDGAKKAVGGLGKLGKDAVEDLESVGKGAVHDVKDVLDSV
GSNGHIGFNEPGTPFSSKTHVVELAEETRANARYFPTLEDVP
GSPRTEYEACRVRCQVAEHGVERORRCQVCEKRLREREGRRE
KNYGNVGHCTKKGCSVDWGYAWTNIANNSVMNGLTGGNAGWHN
KSYGNVGHCKKKCWVDWGS AISTIGNNSAANWATGGAAGWKS
LPERKATRGPA AFLTVQEGCDKFCAFCVVPYTRGAEVSRPFAR
RFRPPIRRPPIRPPFYPPFRPPIRPPIFPPIRPPFRPPLGFPF
RSFPSSFVSPKSYTSEIPCKATRDCPYELYETKCVDSLCTYW
SGIVQQQNNLLRAIEAQHLLQLTVWGIKQLQARILAVERYLK
SSLLEKGLDGAKKAVGGLGKLGKDAVEDLESVGKGAVHDVKDV
SSYDFPGDDIPIVQVSLKAIEGEAEWEAKIDELMEAVDASIP
TLEQWAAWLDNVMMQALKPYEGRPSFPAARQFLLKWSFYSSM
VTCDLLSLQIKGIAINDSACAHCCLAMRRKGGSCQGVVCVRN
YTSVITIELSNIKENKNGTDAKVLIKQELDKYKNAVTELQL
AAAGIMVGSYWTAVTYGAVTVMQVVGHKEGLDVMERADPLFL
AIEIVPGQTVLVPTGLAIYVEDPRYA AFILPRSGLGHKHGIVLG
DQFFGEHLLSDFPASTSLSPFYFRPPSFLRAPSWIDTGLSEM
EHLVIACNTATALAMD AIEAKANVPAIGVVEPGASAAAAISKTR
EQLPTQIHLFATNAQAAGMLLQVLPSKQEQNDDFSLSVLTQTI
FIVLLSQFLVVINGSIPCETTADCPVAVPPEYYKCMYKVCVLIR
FTMKKSLLLFFLGTINLSLCEKERNAAAAEKRDGDDTDVEVQK
GLPVDIMVRRKRLRLKKNNSYMKIYPRTLEDLQQLGLLDEEGEI
GTPGCYSYTSPhWGSTHDVWVATNYEEYALLYTAGSKGLGQDF
IGMIFQHFNLVKRSTVLRNVL SGRVGYHPTWKMI LGLFPKEDKV
KNEFITLNDGRRTLN I KVSNTGDRPVQVGSYHYHFFEVNRYLEFD
MLAEIKGLNEECGVFGIWGHEEAPQITYYGLHSLQHRGQEGAGN
MLAKIKAMIKKFPNPYT LA AKLT TYEINWYKQYGRYPWERPVA
NDTEYTDCLQHSDCQAYACELPFKPDCLMVEYAPQFFRLACGCV
QFGEKFKLLTQSKNKDALSKFVTTSFRTPLSQRNLET LA I IAYN
QSLDGETIELLMVDYGIEAKQKVEVDNADIERFFVEDGYLNEDE
RRIGFDVVTLSGIRGPLSRIGSEPLPGKRECRVGQYVVDLTSFE
TKLAADDLRANYEAELSLRQLVEADANGLKQILDALTL SKADLE

TLTVQARQLLSGIVQQQNNLLRAIEAQQHLLQLTVWGIKQLQAR
TPELFRSRSPGRKKGSKRHKPGSYVIALGKPGVKKSPYMEAL
VFAYGAISILNSLGVNWWVFGSEAGDISLLKKIAMHLAFEDEF
VIEDTYGQHRVKLPAGHLVLYPASSLHCVTPTVTHGVRQASFLWI
ENTIPGFTRPNHGYLESWARQGVLLLNTVLTVRAGQAHSHASLGW
FGCGSSREHAPIAIIKAAGVSCVIAKSFARIFYRNAFNMGLPIFES
FPMKKSLLLLIFFLGTINLSFCEEERNAEEEEKRDGDDEMDVEVQKR
GIFSKLAGKKIKNLLISGLKNIGKEVGMDVVRTGIDIAGCKIKGEC
KVHVFKDGKMGMENKFGKSMNMPGKVMETRDGTIIMKGNEIFR
LAIRWILAACRKRVRGKSMISKMTNEFLDASKNIGNAIRKKDEIAK
LDGLLAGTAAIAFGAYAVLAWNQGQYDVAIFSVAVVGAVLGFLVF
MARTRSRASVTGKMPPAMILTYGSVLGIAGLAMLYSLNPLTAFLG
NVADSRAPRDGRFIETVGTYNPLVAENQITIKEDRVLEWLSKGAQ
PVTGSSVTESTPTKAKSGGCGCDTSTTVEMDEKLQERIAKHPCYS
RMKQIEDKIEEIESKQKKIENEIARIKKLLQLTVWGIKQLQARIL
ADGTLWTIPITLDVDEAFANQIKPDTRIALFQDDEIPIAILTVQDV
AGRIILLRIVTLYLYHVLFSDIKYLQVTCGLILPVKPLPKKTKNM
DNEEKYLRLYAEFENYKRRIQKENQINATYKAQGVLTDLPSIDNI
EEFPTISLLGIDFNLSNIMITVTCVIVLLIAIICTRNLQRRPTGK
FLRRSVSGFQECHSKGGYCYRYCPRPHRRLGSCYPYAANCCRRRR
GIFSKLAGKKLKNLLISGLKNVGKEVGMDVVRTGIDIAGCKIKGEC
GIFTKINKKKAKTGVFNIKTIGKEAGMDVIRAGIDTISCKIKGEC
HGPDSNCNHDRGLCRVGNCPGEYLAKYCFEPVILCCKPLSPTPTKT
KEDVAMARREIISAAEHFSMIRASRNKNTALNGAVPGPPNLPQT
KSETDENLKIAIASSPAPRYITDMSIEEISREWYMLMPRQKITGGL
LDNVAYRLGFATSRDEARQLVRHGHFTLNGRKVNIPSLQVKAGDVL
LEGCGEVLGDSCLSRGYKQAVRLVEALYRDFGVPSTLYFTGNRGFH
LNESREGTMLDFIDAFCEFRPKNIDTLETLLTDNRIWKQRTVGIGV
LNGGGKDKCFRSDCPKHMCPSSLVAKCINRLCRCRPELQVQLNP
LNLNIPYITLYTFSENWQRSSTEVDFLIKLLSIYKTELNNLHKN
MLQKFRIAKEKNKLKLLKLLKHASYCLERNNNPELLRAVAELLKVS
NLLGDDLIDFCANTKVKFIRKHTORLLMESEDSCFISHRMNNSRLF
PQDGRHLMKLVMDGMALQPCNPNFVQARDAILDADRALTNSANKC
QHSLPSIVRYTQEGIEVGYVAAMSSAQDPKNTIVSVKRFMGRSLTD
SPALSIGLSVTLGHVLVGIYFTGCSMNPARSFGPAVVMNRFSPSHWV
VIYTMLAGEPPFNGETAEDIFESILRGNLRFPPKKFGSVSSEAKDL
VYFDMTVGDKAAGRIVMELYADTVPETAEENFRALCTGERGIGKSGK
YRAPLSYPYRDGLLNKDLATDGKLAGARAINVIEKQVGADDLAAVI
YTSVITIELSNIKENKCNAGAKVLIKQELDKYKNAVTELQLLMQST
YTSVITIELSNIKENKCNGTDAKVLIKQELDKYKNAVTELQLLMQ
YYPQIAPCVIVAIRRGDEILLAQHVRRHGGINTVLAGFVEVGETLE
AAGYDLRTEEDCEHTFAALGKIVGFQYLRGMHLNDAKSEFNRSVRDH
ADGSTEQIIFSSAGDVNVYDLQALCDKVGWPRRPLTKIAASLRNSYL
ASRNNKEIREINRVYREELKRDIAKDIASDTSGDFQKALLSLAKGDR
CPLFPYFQDQDQDRAFRVIPADFIIESEGTGVVHMAPAFGEADFFAC
EDYYYIECQRDFDCPQLNSEIFAFKICIEKLCKLEFIYQQAPFLLGQV
GTQRWCNLYGKCRHRCSSKKERVYVYCLNNKMCCVKPKYQPKKWWPF
KGVGIDSIEVERVKKIVAKGDSFAEKVLTPEFAQYQKMKGKRKVEY
LKLAKGFRGRCGTNYKAAKDAVSKALAHSYVARRDRKGSMRRLWISR
MDKIHKFIYALIFFLALFLVNNARNGCIVDPRCPYQCCRRPLYCRRR
NFLTKSTIILTTLFLINAIVLANLSSKKKSDLVSKINEIEENQAENS
RLYLLTRNEDLWRMVCQNAWGSETTRALETVPAKRLGWRLARELT
SSLLEKGLDGAKKAVGGLGKLKDAVEDLESVGKGAHVHDVKDVLDSV
VDEIHLAGFTIKQLDKGEIWDTHSRPVSTEVWKLYQHWIEQYGRPH
WDLHALAHDFETHSDKEEISRKIFSAHFGLAVVCVWLSGMFWHGA
ARLVFVNPEKPCVTDADCDRYRHESAIYSDFCKDGYCFIDYHHPYP
ATCAGQDQPKETCDCCGERGECVCGGPCICRQGYFWIAWYKLANCKK
DLRDAALHAKGKHVAEGVTAIVVPGSLVKEQAEAEGLDKIFIEAGFE
ENIDSLVLASDLILKKWRDYSDETLDILSYSNSEMHNTITPMVRKEDG
FISNDECPSEYYHCRCLKNADEHAIRYCADFSICCKLKIIIEIDGQKKW
GALLITPYYNKPTQEGLYLHFKAIASEVSMPPFIVYNVPSRTGTNICPE
KGMTIKMMNTNKKFEVTEVGVMAPGQTELSAGDVGYIAASIKDIR
KIVKHCRDAHPASASGALLGMDLGGTLEISNVFALPNPGRSNSEERDEE
KKTARSILKNMDSYNNNSNRGIIIREINEKYQMCSKPEYFDYVSKLYQ
LKIKELKVQNKIESPTINWIDGVGIKQDKANPFNYRFEVDIKYTGYYQ
LWRHYFQNTQGLIFVVDSDNRERVNEAREELMRMLAEDELDAVLLVF
SNESRILLMLAERPNEVLTRNELHEFVWREQGFEVDDSSLTQAISTLR
VQAVRSAYKKTIVKLSPNVTDIAEMARAAEANGADSVSLINTLLGMA
AELFREMAAWGHPRAAEGDPEQVQGAEGSSEPAAPAPEPFTVHFLSLD

AHHFLDFPIENIKVVIHPQSFVHGILELIDNSFLMHTSQTDMKIPIMHA
 ALLTNLESQDVLFIIDEIHRSPVVEEILYPAMEDYQLDIMIGEGPAARS
 CCGGRMKQIEDKIEEIESKQKKIENEIARIKKLLQLTVWGIKALAAIA
 CCGGRMKQIEDKIEEIESKQKKIENEIARIKKLLQLTVWGIKQLQARIL
 CRDDRIREYFYGFRGCFYPHAFDVKFSDVKIYKVGAPTIPDSCLPLGMS
 DLFSAHELLPGPLSPMHAQLSPPSVDSSGLQLSFTESPWETMEWLDLTP
 EVDWIYHLCDDTDTCPEHWSKFFIYKCVNHVCDSSISKVTTDSKEYKNFP
 GFLLEDCEWLITEKDCLLYTEVLYVGISGMTYSQTGILIERAFATLY
 KHEAVQTRFPPEPNGYLHIGHAKSICLNFLAYIYDGLCNLRFDDTNPE
 KQVIGHLAQASTDYQHPFLVEREQVKRVGFIVISSDRGLAGGLNNNLFR
 MLTIGGIGFILASISSYTGRDLLPLGNPSSLLFIPQGIIMGAYGVIANL
 PLTDVWFAPPLKTYQSQLIGQHVTHVETR GKALLTHFSNDLTLYSHNQ
 RYLDYSVKILKGIENPNIFSWNVITIRGFSEENPKESFLLYKQMLRHGC
 TAEIKSTDDELHLENGEALEISQLKESSSGQPYDAVFPLHGPNGEDGT
 APRSNRWRCTISQPEWFKRRWQWRMKLKGPSITCVRRFALECEIRAI
 AQSRQORDAEFEKQREQIEALNADPYDIEAQKSIETAIRMEAVLENMQHAME
 DGFKDVNDTYSHKSGDAVLKQMSQLLKNYVPNQFKIFRNGGEEFSVVIHN
 ELCEKASKTWSGNCNGNTKHCDQCKSWEGAAHGACHVRNGKHKMCFCYFNC
 ELCEKASKTWTGNCNGNTKHCDNQCRSWEGAAHGACHKRSWKWKCFCYFNC
 EPDEICRARMTNKEFTYKSNVCNNCGDQVAAACEACFRNDVYTACHEAQK
 HPGTFHVRIKCMPPKMTAVFGDNCSFYSSMGDLNNNTKSVCCMVPVRMDNI
 KLCERSSGTWSGVCNNNNACKNQ CIRLEGAQHGSCNYRFP AHKICICYFPC
 KLCERSSGTWSGVCNNNNACKNQ CIRLEGAQHGSCNYRFP AHKICICYFPC
 KNIDGRVSYNSFIALPVCQTAADCEPGRTRGRYKICINNKCYPKLLKPIQ
 LKEEGYIKNYNIFDKKGISFIKAILNYDNKRNPAINRIDAISTPGRKVYS
 NLCEKASKTWSGNCNGNTKHCDQCRNWEAGAKHGACHVRNGKWKCFCYFNC
 NLCEKASKTWSGNCNGNTKHCDNQCKSWEGAKHGACHVRNGKHKCFCYFNC
 NLCEKASKTWSGNCNGNTKHCDNQCRNWEAGAAHGACHVRNGKHKCFCYFNC
 NLCEKASKTWSGNCNGNTKHCDNQCRSWEGAAHGACHVRSWKHKCFCYFNC
 NLCEKASKTWTGNCNGNTKHCDNQCKSWEGAKHGACHKRSWKWKCFCYFNC
 NLCEKASKTWTGNCNGNTKHCDNQCKSWEGAKHGACHVRNGKWKCFCYFNC
 NLCEKASKTWTGNCNGNTKHCDNQCKSWEGAKHGACHVRNGKHKCFCYFNC
 NLCEKASKTWTGNCNGNTKHCDTQCKSWEGAAHGACHKRSWKWKCFCYFNC
 NLCEKASKTWTGNCNGNTKHCDTQCKSWEGAKHGACHVRNGKHKCFCYFNC
 NLCEKASKTWTGNCNGNTKHCDTQCKSWEGAKHGACHVRNGKHKCFCYFNC
 NLCEKASKTWTGNCNGNTKHCDNQCRNWEAGAKHGACHKRNWKWKCFCYFNC
 NLCEKASKTWSGNCNGNTKHCDNQCRSWEGAAHGACHKRNWKWKCFCYFNC
 NLCEKASKTWSGNCNGNTKHCDNQCRSWEGAKHGACHVRNGKHKCFCYFNC
 NLCEKASKTWSGNCNGNTKHCDTQCKNWEAGAAHGACHVRNGKHKCFCYFNC
 NLCEKASKTWSGNCNGNTKHCDTQCKNWEAGAKHGACHVRNGKHKCFCYFNC
 NLCEKASKTWSGNCNGNTKHCDTQCKNWEAGAKHGACHVRNGKWKCFCYFNC
 NLCEKASKTWSGNCNGNTKHCDTQCRNWEAGAKHGACHVRNGKWKCFCYFNC
 NLCEKASKTWTGNCNGNTKHCDNQCKNWEAGAKHGACHVRSWKHKCFCYFNC
 NLCEKASKTWTGNCNGNTKHCDNQCRNWEAGAAHGACHVRSWKHKCFCYFNC
 NLCEKASKTWTGNCNGNTKHCDTQCKSWEGAAHGACHVRSWKHKCFCYFNC
 NLCEKASKTWTGNCNGNTKHCDTQCKSWEGAKHGACHVRNGKHKCFCYFNC
 NLCEKASKTWTGNCNGNTKHCDTQCRSWEGAAHGACHVRSWKHKCFCYFNC
 NLPMQEDAETWLQKTTVEHPDSVEGIKNQNVCYGCSVSLKSLKGLTHIPD
 PRHLFVQEQALQAQAYEDHPLPIGYGQTISQPFIVALMSQILRVTPGMRVL
 RVTLSKPKQALATPNKEEHGKRKKKGKGLKKRKLRLRKYKDFCIHGECK
 SEQACPSLKIEYAFRIYDFNENGFIDEEDLQRIRLRLNSDDMSDELLTD
 SQKQFLGQVVISILFYLVYHFNDFEAETLNPFTNIEVDLGFVVFIFILFW
 VVENIEYLQDVIENLGDKNNVIMPGYTHLQRAQVVTFKHHIMAYYHMFKR
 YPASMDNYDDALEELDNLDDLDLDLDLDLEPADLVLLDMWANMLESDFDDME
 YPASYDDDFDALDLDLDLDLDLDLDLEPADLVLLDMWANMLDSQDFEDFE
 ALERTRTTMTLSYNIGCKSDADCPKAIEPHYTRCVDGHCWLYFGGPKLHN
 AVGSLKSIGYEAELDHCHTNGGYCVRAICPPSARRPGSCFPEKNPCCKYMK
 AVYANWEFAKIRGIGWGAGVIWLYSIVTYFPLDVFKFAIRYILSGKAWLN
 DDSIQCFQKNNTCHTNQCPYFQDEIGTCYDKRGKCCQKRLHHRVPRKKKV
 DEEKRENEDEENQEDDEQSEMRRGLRSKIKEAAKTAGKMALGFVNDMAGEQ
 EEEKRENEDEENQEDDEQSEMRRGLWSKIKEAAKTAGKMAMGFVNDMVGEQ
 ELCEKASKTWSGNCNGNTKHCDQCKSWEGAAHGACHVRNGKHKMCFCYFNC
 ESKKYMEEAAKALGFDVKIHADEIDPLGGLELAIDENAI SADHLVASSTEGK
 FKAFAKFAKFAKFAKFAKFAKFAKFAKFAKFAKFAKFAKFAKFAKFAKFA
 FQLAAASWISGFSNSVLQSTWTLKMPLCGHKEVDHFFCEVPALLKLSCVDT
 GKKS VGAGNAIAVQNLGENSAMLMLGLYSLAVLVGVPVAIGAIGFGVLFA
 GLEYSQSFPNGGEFAVCETCLRLGRGKCRRTCLDSEKIA GKCKLNFCCRERI
 GLEYSQSFPNGGEIAVCETCLRLGRGKCRRTCLDSEKIA GKCKLNFCCRERI
 GSGYGGGGGSGAGSGNRVYVGNLSGWVDDMALESFSEQGVVEARVIYDR
 KFEICERPNGSCRDFCLETEIHVGRCLNSOPCCLPLGHOPRIESTTPKKK

[illegible]

DRYAEQIPVTFVLGFYVTLVVRWNQFVNLWPDRMLFISSSVHGSDEHGRL
GGDDAATSGPPPSLEWRFSQVFGERTAGEEVQEVDIIISAEFDKSGDHLATGDR
HKIKHYEYEGNVGVEHALLPEKGI VVP GDVIIGADSHCTY GALGAFSTGGGST
HLIKKKEALLVPLGGAMQEA AEKVRICSCCGNVDTSDPCTICTDERRAPATLIV
HWCAALEIDGIGDKLIARLVSLGLVHTVADLYELSAEQLAGLERLGARSAKIV
IALIALDRCVCVLHPVWTONHRTVSLAKKVIIGPWVMA LLLTLPVIIRVTTVPG
IDVPLEPRAYPQGLCKSHWGLNEVPCVESYKGLIFGNWDT SAPGLRDYLGDI AW
IISLGRAEEFISALSLLIQQLVVDHLHIVGDVYDRGPYPDKIMDTLMNYHSLDF
KPLILKACANMFSQYMC SLRFDYDDVDFQQIVQYFDEIFWEINQGHPLDFLPWL
KQSKAQIVPFSRKN AEGARAQLIDGWLTFDGD RIMKASEMQIPGLHNIENALAA
KRRGHAPGSRAAGNTTSGSKGQNA PKDPRIGSKTPIPLGVAEKVTKQHKPKSE
LADKGYDGMADVWSCGILFVLMAGYLPFDDPNLMTLYKLICKAKVSCPHWFS
LKL VYIAIIYVSFFLVVCEGEKCVTADDCQGHMPAGYHFICMNARCVLVYYN
NPLAEMITLAHQHGAKVLVDGAQAVMHHPVDVQALDCDFYVFSGHKLYGPTGIG
PAIFLMGPTAAGKTDLAIELTKVLPCELISVDSALVYRGMDIGSAKPSKEILAA
PRMLDLNMLLFS TSDFCIGIVALPLTMVIVSGGIDISFGSTIGLCAIALGVLF
RSPNDDDSMGIQIVKDLHRTGCSYCGQEA EQDRVVLKRVLLAYARWNKNVGYC
SKYSDEHVESILTEIAAVLNKHNASPELTMVVGN IATNVINQNVA AQRKVIA
SRSLAACEGAILVVDAAQGV EAQTLAVCYTAIDQSLTVLPVLNKIDLPAE PER
TNDTIESLGIERTFQKLDQAEIVLWMVDSSDASSQIKQLSEKII PRCEEKQLIV
TVVTD AQLVEQIKSKIYQSHNNNNNNNGGTTTTTTGT VNLNQQIIDRVTKGI
VFAKAGSGNIVVNQRPLDQYFGRETARMVVRQPLELVEMTDKLDIYVTVKGGGI
VGFLMMSHMTTPENLAVEAKKMESYGATTIYVVDSSGALSMDDVRARFRALKAV
VYILILFICIFLVMIVCDSAYLPLSRSCITDKDCSRVKNYNARCRKGYCQYLQY
YWRGSEKNPMLTRVYGTAF AKKEDLENY LKFL EEAQKRDRHKLGPQLELFMLNT
ALIKIYAAKNIEVVQGYALTESCGGGTLLLS EDALRKAGSAGRATMFTDVA VRGD
DNGAGYALVTPPYIRPNQGG LKAHYLAVAEQGLPVVLYNVP SRTGCDLLPETV
GLDTSVFS TKGATYITYVNFNLERVKTKPEGNSHGIPSLRKSSDDPGSSFV VAG
GSGPAGLSAAIYAARASLNPI LINGIQPGGQLTITTDVENYPGFAESVQGPWME
HIIGDVAGRDCVLVDDMIDTGGTLC KAAEALKERGA KRVFAYATHPIFSGNAVDN
IVVSGEMMAMYAANNIAKIGTVKYASSGNVRLAGLICNSRNV DREDELIEALAAAL
KQATVGDINTERPGILD LK GKAKWD A WNG LKGT SKEDAMKAYINKVEELKKKYGI
MLIEEDLGGFLVKLIKEVKSVDVETGLETVYRSAYAASKDLGKYSTS VRNACAKK
MNNLNKFSTLGKSSLSQIEGGSVPTSVYTLGIKILWSAYKHKRTIEKSFNKG FYH
MSKSQSLQDPYLNALRKERVPSIYLVNGIKLQGQIESF DAFVILLRNNISQMVY
NGLLIGDDEHCWSDNGVSNIEGGCYAKCIDLAQEKEPDIWNAIKFGTVLENVVFD
NHAISGLLPKLPFGCCTSNLDCPRH MCTHPQQPWCIFYGNRIMYRGSR LGICKCS
PNKTGNVIDWDTK EIVGQHQAHHYYTIGQRRGLDLGGSEKPCYVVGNIEENSIY
SKDKRIAFVTGHPEYDAQTLAQEYFRDVEAGLDPDVPYNYFPHNDPQNTPRASWR
VVAPYLQGERFIDVGTGPGLP GPILSIVRPEAHFTLLDSL GKRV RFLRQVQHELK
YDYIGNNPAKGGLFRAGSMDNGDGI AVGWLGHVPFRDKEGRELFVRRMPTFFETF
GIAANGFIVLVLGREWLRYGRLLPLDMILLSLGA FRFYLQLVGMGHN FYHSAHVVE
ILRKDFIIDRIQIEESVRIGADA ILLIGEALPAKTL YELYQEAYDKGLECLVEVHQ
LDKTGTITYGNRIASEFLPVNQQM LEKLIVAAYMSSIYDDTPEGKSIVRLAKQMYI
MAGNLKIVYALMILVSLILVVTSHSFLPCVT KDDCAYDECISPRKPTCYLETCHCL
MAHILKFVYATILVLF LFFVATKVDGAVHKECKTDVDCRQIWFVTKCINHECQPI L
MGQIEKFVYSLIIILSLALVVT CNIGPICQTYMDCPSDMCTRPKHAYCVSYKCYCV
MKKFNVQITYTGMIEETIEAESLEEAEFEADV IARLEAPFDCDEYEINVEEEQEND
MYKLTVFLMFIAFVIIAEAEICIGGGGGCSVFSGPSCCGGTCKCKFVLIFPKGCHCT
PTSYGRTDVVYNLQRLALLTTALGQTPINPHLVYEV MKDKVHQPYRASLIPGLQNI
QAYVKLQVAAGMANPSPVP GPALGQQGVNIMEFCKAFNAKTDSIEKGLPIPVVITV
RVC PGIVIVQHMEKF TA AFAARLNGLCQI AVKEAANNDRVMPGRALIAPGGKHLL
VAVTINNRIKKLQYERIRKLEKRKRGEPEFIFNGNYSLEEIELFLHFRKKSEGKK
ELAKVQLQTGCASRQRATRHTLLRVLEAALRLLHPIIPFITEELWQTVAPMCDAKTA
EQCVSDADCQIKFPGPRQHLLRCTQGNCVMLVGQ GKNYFSIMSKTLFSLLVIIIFLLL
GLENYDGGRAGDP PATDLARRLRDLGFKVARLKTGT PPRIDRRSVDFSRMAEQPGDD
GTAKVMAISNKETT VKTGP IQFTVKKVIAAVANEQLPFIDVQIESENTSDEVVRFRP
GVIPGQKQCI ALKGVC RDKLCSTLDDTIGICNEGKKCCRRWILEPYPTVPVKGKSP
IGPASVRLFGEVLWDAQTILWNGPMGVFEMDKFSRGT FKISHEIARSHGIKVVG GGD
ISIYVRCASTNECYTTFKFAPLGS MRCVEGYCKHLKDFKVKTPLQIKEITPLLLHFP
KMEILLGSNGGARCDINKKSFDCYHRNGRCR FNCRKREYNNGD CSQYQSCCLPTRNL
LEELLAPPAARGEKETAEENARLRQLIRRMQIRLLALEKSYHSRRAMQSRYLQQIQ
MAKIIKIVYVMIVFFFI FLSVTNSSAFSGCMNDSDCPDLFCLPPLDMKCHELVCKCR
MAKVYMFVYALII FVSPFLLATFRTRL PCEKDDDCPEAF LPPVMKCVNRFCQYEILE
MAQILKFVDALILFLSLFFILINGDRIPCATDADGPPKILPIIHKCINNFC KLKLYN
MKQYKVLNEKEMKKPIGGESVFSKIGNAVGPAAYWILKGLGNMSDVNQADRINRKKH
MRIRKHANKIIGWLFIFLAVLTIVKVIVLGKPLDGF DIVLYVLLQLLYGMEKLSNDE
VDRYDAKEEARKKYLKEQQLKKLEEKNNNENGDDATSDGEEDLDDL RVDEAKVDES R
VLGFPFIEGASVAAEILDHKG RPKVIAFKKRRRKN SKRKRGRYDEITVIRVTEILT D

VSKIKMNEYRGKPPYSMDDLAEDPQLERRYIQRERNHVKERDLDRRNKRQAKRNGY
GQGANPIACILSFAMALRYSFDMGEEATRLEKAVETVLADGVRTADLMGPEGGTPVS
GVLIGSISLTNEPLKLITALILPHGIFEISAMLISAVAGFKIPYKVTLYLLDKKEKPL
IAAHKLGAKSVLATDISDESMTAAEENALNGIHDIALQKTSLLADVDGKFDLIVANI
KLTGCEVDGDCPKVFKLVKMILFIKCINNKCVRGLLSQTGTQCPDFFFLKRTLPRFYF
KPRQEKTYREKARHLGASPOKRGVCTRVTTPPKPNLSALRKVAKVRLTNGFEVIGYI
LGVNKSASKDEIKKAYRKLSKKYHPDINKEEGADEKFEISEAYEVLSDENKRANYDQ
LILFISLFLDVVKGHDOQRECYTNYDCCVKYSCPYKHMVKCVGGYCLGFRNDYGKKNLY
LKDKGIGKGKTREDHADTMNQLFSAYAOGKQAKELSVILGESALSDTDKLYAKFADAF
LLETTQGKTHPISPVTALWIIAIPMLDMVAIMYRRLRKGMSPPSPDRQHIHHLIMRAG
MAEIFKFVYSVILFVSLYLFIYAEKECDTDADCRKKFAGANQHLLWCNNGYCECHTH
MAEILKFVYNATLFFSLYLVVYNSKLWCDDADCEKFPGPSKYPIKCMKGICKCVIN
MAGILKFFYIAIIVVSLYLVEGKDGCKTNFDCLIKYPDHNEIDILQCIGGHCLCLTN
MAKIFKFIYGLVIFLYLFLIQKEVAGYIQCDFDADCEMFRHIFYLCIDKLCRQFVTL
MAKIITIIQIFTIIMLFIFVIVTDASYPCIKHRDCTTITCSYPLVPRCLIKQCYCGFN
MAQSLIFVYALIIIFLFLFRVEAEHLKIRCVTDDDCPKVEKPLYMYCGNHWCAKLFHV
MDEVFKFVYVMIIFPFLILDVATNAEKIRRCFNDAHCPPDMCTLGVIPKCSRFTICIC
MGGIIKIVYAFVIFISLILIVTSNVHSLPCGTDDDCANDPCIHPEYPHCHEQCHCV
MKKLKVMNNGELEKVIGGSLEYEMKNSVPRLLGPDGMEGSMGGSTGGIQSFRHFPGFGR
MKTFSVAVTVAVVLVFCIQSSATFPEMPYNRQKRGFKCKFCCGCCGAGVCGMCKKF
MNNLTAQDLLDELLELKDQGHAEEDYIVVGEPINPYDHYLEFDGWHLDSNTNMIILKG
MNNMVETCKYFYVVLFLFIFIMATDGVYLCEDDEDCHIMPCMVPEYAKCIRMICQCC
MNTILKFIFVVLFLSLIFLSAGNSKSYGPTTLQDCETHNWFVESCIDFECKCWSLL
MRGSHHHHHHAIDVIEGRWQEWQKITALLEQAQIQQEKNEYELQKLDKASLWEWFG
NKSFSEDKLTAKISVNDFIILAVAKALQEVNANASWAEDAIRYNNVDISVAVAIENG
NVEISMERIKKDFYSIKNKYDYITVEGSGGIVCPISTGKEKIMLDNIIKIFKLPATIV
PKASFDGDKDFLEKIRKALYMSKICSYAQGFAQMRKASEDNEWNLKLGDLAMIWREG
SFVIACGAVCTPQILWNSNIRPYALGRYLSEQSMFTFCQIVLKRGIWDAIATDPRFAAK
SVVKDLSLIVDAAVGFNGRCLPAGPLREPVEAGLARADLLLSIGGPEAQRFFATDW
TAMHGSNTISAAAPFFMTQNSVANLTYRILYFLKPLCLPPFVAQILCGLLFSPTVLG
TFLKKTLDLMEGLDYEILTNLFIETIYRILYFLGSLNFHECAICHRDPLDLSHRFSA
VKRLYSFAKPNFTFYIRVTPAIALERIRKAKRKIKPQEAGIDILGEIPLEEGFLKYQ
AKKPFVQRVKNASKAYNKLKGLAMQSQYGCPIISNMCEDHCRKKMEGQCDLLDCVCS
DDNGIGVSAVQKLVELDSYIPEPVRAIDQPFLMPIEDVFSISGRGTVVTGRVERGIIK
ELEPEDEARPGSFDRPLAENNVVRTIIEFLTFLHLKDAGALERLPSLPTAESAEADERS
HASAEVISGTRRMVVPVASNVAARLTAGATVMLNEKLVLVEQRDADTVGQIRSVKQVL
MGDIKVVVFAMIIYLYMLTIVTNAVITICSDSDQDCRRYRCDPPEYPRCLGILCKCVYVSG
MSKTIMFLYAMTLFLFLLHIEKSSGVLIDCKTVKDCPTSYTKIYRCEDNKCRFSFVIGL
MTKIFKFTHAMTLFLSLFLVAESYFADILCKVHEDCPQKSTHKYCIDDECFLYYWEAP
MTKTLIFIYALFIVVSLFLVVTSETRIPCVRNDPCPKRPYPLFMKCIDNFCEIWKIGKE
MVAMSEELGVDMEIRSLERMLQYGEQNIRRAVPLALGLLCISNPKVTVMDTLSRLSHDT
NIEKSGAWFSYNSTRIGQGRENKQFLRDNPAEIEENAVRANAGLIAEEMIGGPGG
PSSLTSMWVFGSLLGVCLGVQILTGLFLAMHYTSDTATAFNSVTHICRDVNYGWLLRYL
QCSHTRVKETRAALATLTRVNGKRATLHVLGASGTVKKATEKFLKDYTVFEPEIHEKRD
RAWDDFFPGSDRFARPDFRDISKWNVRVSNLLYYQTNYLVAAMMISVVGFSLSPFNMI
SGHRVLGDISLSQLKRPSLKHQPLNKKANAQYLFAERSCFTLMDTIIALGFRRVLCVG
SQILMFVSLIIFLSLFLADTKHTENIPCNKRDCPQPLYPKFVTCFEGLCRMHYPLKKI
VSIPAGQPEKTALNLTGPILINHRARIGLQVPQTDASLPPQFYLMDDANGSTTVRRKA
AAGKKLKETGLKFDIAYTSALQRAQKTAQNILEQMEQSDLELIKTPALNERNYGDLSGLN
ARVSKVTKGGRQFSFRVAALVGDKKGVGYATAKHDEVLDARSAVRKAKGAMIRFPLRE
FEIKKPTKFKSNINLNLKSENVKAGTEFRKASRIGGSYNVRSNIPLTLIDEIYKRIEEL
HMRVRKKIFGTAEPRLSVYRSEKHIYAQLIDDVEGKTLVAASSSEKGFVGVSNGEGAK
IVIDVGINRQTDGTLIGDVEDYATAEKRAGWITPVPGGVGPMTVASLIENTLEAYVKYHSN
MADTLKFVHVLILLISIFLVIIVSSFIPLPCITDKDCQTLKKNKGKGRCKRGFCVDGLIG
MAEIIKFVYTMILLLSLFLVTTKVGAYIACQSEIDCPPNYSFLFAIRCIKQKCVTVGRYL
MAQILMFVYFLIIFLSLFLVESIKIFTEHRCRTDADCPARELPEYLKQCGGMCRLLIKD
MDQISKSFYALMIFLSLILVVTSTNDIKCTVAGDCPDFFRCPNTFVRCISNICICRSLSH
MNKNMPQILMFVYTFIIFFSPPFVVTNGTTSCTIDDDCPKAVSFLVFKCIDNICVRVEIL
MSISVGDKVFNPETNSTLEIVQLVGDIRDTHYKLSDGSIISLIDFVVKPIHLIKEEQEND
MTIIIKFVYIMILLFFPFLVVSQIFPKWCLYDKDCPQNMCRPGRIKPCIFGHCNCVKQRS
MVKTFFKFIYSIIIFLSPFLVVMNVDTGELIKCTMDADCPSTLNRKWLINNCRKMCVTNV
PLTDGFPLDILVANVYACFLLGTVDTALYARKIHSRDVHTIIGTGMMGGVSTFSSFAYGSV
QPRATRKGVTPKQGYCEPFLLDGCPFVLLPVCSRDKGCKGKTKCCFYQCQMRCEVPWTTLT
QPVGINTSTTCCYRFINKKIPKQRLSYRRTTSSHCPREAVIFKTKLDKEICADPTQKWV
VLGLVATGVGSIQGHVNGSLELTLAGVATVIGGYKLQKRLRQKYTLEGIKDSIKIKMD
YKVLNDYEQRQKQLMVENAELKKVLQOMKKEMISILSRKTKELDDSIGPVASDIEED
AEYGDLTRGPRIITEETRAEMRKILHEIQTGQFAREFVLENMAGKPGFTAMRRREAHPIE
AGIRVTAEVSPHHLILDEEAIPGNDGNWKMNPPLRSKEDRAALLEGLDGTIDFIATDHAP
FGLMIADPICSILIAILIVVSVIPLLRRESVGILMQRTPLLENSLPQCYQRVQQLQGVYSL

LIDHLGAPRWAVDTILGAI VGNLASWVLALVPGPGWAVKAGLATAAAIVKHQGKAAAAAW
MANYYYLKRLLDDGSGKLHQVKCYKSLDKAIKYAKEMSTFKKTYQVAGQFDMNRLANTGRLI
MFTLKKSLLLLLFFLATINLSFCEQERNAEEERRDEPDERNAEVEKRFFPIVGKLLFGLLGK
MKFTATFLVLFMVVLAEPGEGFLGILFHGVHGRKKALHMNSERRSYDERQQQQQELDKR
MKFTATFLVLSLVVLAEPGECFLGALIKGAIHGGRFIHGMIQNHGDEQELNKRANDE
MRINRTPAIFKFVYTIYIYFLLRVVAKDLPFNICEKDEDCLEFCAHDKVAKCMLNICFCF
MSHVAPEIEWTVRVISALLLLGLVLAETASWIVGPSRGMVTAQKNLLPAFAKMNKNGVPV
MVKTPKLVYVLIIFLSIFLSMIVSNSSFLGTFISSCKRDKDCPKLYGANFRCKRGTCVPPI
QVTPVTCKWGSGRCRTTCKKSEVYIILCKTEAKCCVDPKYVPVRSKLTDTNTSLESTSAV
TKLVKENDDAGEFWSGSVYSDQYRDQQLVGYPSSLVTELTVYKDELQELPTTAVDFYCD
AFLEGSPEKAALTNELLAFAADPAITQGPEYYDLTKAEQREQTVKKIARLALYLENDIKLA
ASIPPDEKVVLLGHSFGGMSLGLAMETYPEKISVAVFMSAMMPDPNHSLTYPFEKYNEKCPA
GDVPPGIRNTICLMQQGTCLFFCHSGEKKRDICSDPWNRCCVSNRDEEGKEKPKTDGRSGI
KKDVGKEFVNVLVKDDGVSKNNLLMQFQADILGIRIVRPRVIETTSMGASMLAGLAVDYWSS
KKSLEDFKEASGMSDFIQQIILPSENVMEVKKGEHKIVEKYIWPGYLLVKMHLTDESWSYVK
LNPSNLPDGDNGGGGVYEFGLTKSSRTSLTISNDVYFDLGSQRGSGANANRGTINKIIGVRK
MAKIIKFVYVLAIFFSFLVAKNVNGWTCVEDSDCPANICQPPMQRMCFYGEACVRSKFCT
MAQLIIFVYALIIFFLYLLFVEAQITKLPCVTVDPCPKVEKPIPMVAKCFGKSFSRCHYFYF
MARTLKFFVYAVILFSLFLVAKGDDVKIKCVVAANCPDLMPVLYKCLNGICVQFTLTFFPV
MQRKNMAQILLFAYVFIISISLFLVVTNGVKIPCVKDTDCPTLPCPLYSKVDGFCMLSI
MTKILKCVYAMILFLPLFVAMEVGRRANVECESDKDCQEHWSEFFIIQCIDNICVPSEPL
MVETPKFVYNLILLIYIFLFIICDSTYLPTRITCITDKDCPSVKNYIGRCRKGVCQASKLR
MVQLLKFFVYAMILFISIVFLIRTQLSDIYECECTDDYCPKYRDLVYFKCIDKRCELVEAHA
NALRRFQQLIDAGYTTTVRTTRGDDIDAACQLAGQVNDRTKRSQRYKIDKNALDASNADI
NRWYCNSAAGVGGAAVCGLAGYVGEAKENIAGEVRKGWGMAGGFTHNKACKSFPGSGWASG
SVTLQPGPEDTGKACGVDFEIRAFCAKSIEEKIHKRNSVRLVIRKVQYAPEKPGPQPMVETT
SVYKSDIHLPAYASFVIDNGVVRDISVNDLCFANKFFQFDQWYESTFGSVYYHNSMDCPIV
SYAKGAKELSQAAKEPAHALVLGIPTLQTVFGGIIMGALAAWCYNKFYNITLPPFLGFFAG
SYFGEFGKQIGYDVEKLLHHIERILGLNQHWPVVLVGIGLLGQAIARYEGFRSEIEIVAL
TTFHPILVILFSLFPFIPAPISFACNSRVSDCPKIRCINIGFVLRCLYNQCKVRITQLVDYVLK
DPLIKAITGEPDIDLSEIMINRANCSEIEMVMDGCLNDEHLEELGGILKAKLEGHFKNQELRQV
GTEVPVPGGEASTIGSVHVTREVDAAARTLETHQIAFRESGLEEALSRVIAIVVQPGVEFDHTQ
LGRKRAMLVNNVLAVLGGSLMGLANAAASYEMLILGRFLIGAYSGLTSGLVPMYVGEIAPTHL
MFTLKKSLLLLLFFLGIISLSLCEQERDANDEEDGGEVTKEVVKRSLRGWCWTKSYPPQCLGKR
MFTLKKSLLLLLFFLGTINLSLCEQERDADEEERRDDDEMDVEVEKRFLPLFLPKIICAITKKC
MGTIHKFLYAMILFIHITLVVSGNFFEFFHKCTQSDCPSLLCRNKSELPKCIAGFMCRCPNV
MKQIIITENKVILSKILGGSSSIDDIGLNDTEHMLPLYSKKSNNHNRDVLNPRYQTHFKFM
MRLHLLALLALLFLVLSAWSGFTQGVGNPVSCVRNKGICVPIRCPGSMKQIGTCVGRAVKCCRK
MVEIHNFIYAMILLVFMFIVVVDWSWGLTTECVTELDCYKKYRLPAEKKMKCIRGSCYRVRE
SDCTLRNHDCDDRHSCCRSMFKDVCTCFYPSQAKKELCTCQPKHLKYIEKGLQKAKDYAT
SSNVVLFLSEIMGYFLSSILLIRKSLRNEYRGIITDVLGGDIQDFYHRWFDIAIFVASAFLS
TPMLHWPDSMMTYLTGDAVLFSNDAFGQHYCDEHLFNDEVDQTELFECQRYANILTPFSRL
VFADPGARRRLEAITHPAVRLAMREETARLAAQGHPLAFYDTPLLYEVGLEALLDAVVVVWAP
VIDDNGLTIVYAEPDQFAQVQDALREAGVEEFKVAEFEMLPQTDIELSEEDQAI FEKLIDALE
VLVASITLSLFGFNGFMAANGVDLNLSSLLVFCVAVFGFAGSLVSLFISKWMAKMSTGTQIISQ
ARDAIANDRNPCVYTCALNPYCDSECKKNAGDSGYCQWFGFRGNACWCKNLPDKVPIRIPGECR
DVKAGSAGFMPTQKSTIELAERIVKVGNGAGVATTALITEMSQPLASTAGNSIETREAVRYLKG
EVPQCVRINVDIPEEANQNL SFGATERWWEQTDLTCLIISNNKLQSLTDDLRLLPALTVDLH
IMQKMQPIAPTMDPMQVKMQWMPVIFTVFFLWFPSSLVLYWLVGNIVAI IQKIIYAGLEKKG
IPIVNFDDVLDLVTGVVNI GLTAVGATAGVAIGVVGATAGTAIGVAGAATDAVTNIAKGGAL
IVRSGKTLHELKQGRKYPQRLVNVRLAERVSLASVPLVQKVKAEEAEELGDSGRVLLRPSGTE
LHKGRCPACPEGSSAANGTMECSSPAQCEMSEWSPWGPCSKKQQLCGFRRGSEERTRRVLHAP
LPRLRELLNASDSPRLRVLYAELGEHDSTASLLVKAVAEQPLKLTGGVIASEYDVELDELK
LQVSDHLQVRRLRVTETKIYYLKLHPDHPKTVFHFWRILIQILQKGLSITTKDPRILVTHCLV
MNHISKFYVYALIIIFLSIYLVVLDGLPISCKDHFECRRKINILRCIYRQEKPMCINSICTCVKLL
MRIFSLIVAGLVLLIQLYPAWGTLYRRFLCKKMNGQCEAEFTFEQKIGTCQANFLCCRKRKEH
MRLHLLALLALFFLVLSAGSGFTQGIINHRSCYRNKGVCAPARCPNRMQIGTCHGPPVKCCRK
MSRKYRGFDATAFTKDGVCVYVIKDDKEKGTALVRDTRGGDEYTVSNEGMIYTKDMESIDVMPW
MVETLRLFYIMILFVSLCLVVVDGESKLEQTCSEDFECYIKNPHVPFGHLRCFEGFCQLNGPA
QALISSIEAQLSELRSEMECQNQEYKMLLDIKTRLEQEIATYRSLLLEGQDAKMTGFNTGGNSTT
QGLDPIARLFEMIFPITVIIIFLTIALMSLGI FEINNLRPVLGDGIMPVLRGVKTTNLSFTCSEI
SHICPDESTLCMALLQLQREGHVTVSLHEGELVKFSQAGQGRVSPVSEVDLGIYQLQCSEKLL
TTIRFDFEKEQRLGLIPQREGQYKKNVLT SATVKVNHYSKQGTQSASKAAVAVKANKAAG
DLSQRDPFFILPVLMAVSMYVQTS LNPTPPDPVQAKMMKFMPIAFSVMFFFFPAGLVLYVVVNNV
EDKKLRALKPGEALVQTEYCGVCHTDLHVKNADFGDVTGVTLGHEGIGKVEVAEDVESLKIGDR
ERQIFDAPRFQYISELYGYLAKQFTVFGQHVHIGCPDADSALFLLHMSRFPHPHIALSASSPYV
GKGIKKEIEAIKKEQEAIAKKKIEAIEKEISGIVQQQNNLLRAIEAQHLLQLTVWGKQLQARIL
GLGPAEGHCLNLFVGCRTDVCNIVEDQIGACRRRMKCCRAWWILMPIPTPLIMSDYQEPLKPNLK
GLGPAEGHCLNLSGVCRRDVCKVVEDQIGACRRRMKCCRAWWILMPIPTPLIMSDYQEPLKRLK

GNPRIPHLSSPGQHLSPAPPLSHTGVAEYQPPPYFPPPYQQLAYSQSADHYSHLGEAYAAAMNP
IAYVGTVPGLSPLWRLFMITGLAGGLSTFSTFTAELFSLREGRLGMSAGMLGLHVGGSLALLML
IRSLESMFHAHHAKLEPREKAELARELGLQPRQVAIWQNKRRARWSKQLEHDYAALRSKYDALH
KLLLKPLEGIERPALVTVLPHQKQKTVVLDLGNVDCDSTMLVQFAIMGSVLAEEVVGIPNPRV
KPLRRLTWLMAIIYCGVYTALVRHADASLAYGGWLPFHDLPVHSEQDWVQLTHRIMAFIVFT
LAEQRDELEGFQDDAGRKKHSIILRTQLSVRVHACIEKLYNSSGRDLRRALFSLKQIFQDDKDL
MARILCFFYGLLIFVSLFMVATNQSIDVLPCLFSNECPPDLCPDTLFAKCINLTCQCTAEYDLD
MARILYFFYTLILFVSLFMIAINGSLPDAPPCLFTPECPPDMCPTDLTKCINLTCQCTSEYDID
MKLTEKELNTILRDEDETGNNGTAFLGETLADFLEESGIDFTNLITILEVNELLENNGIRPIEVVPC
MLRLYLVSFYLLKRTLLVSYSFYSTYIIECKTDNDPCISQLKIYAWKCVKNGCHLFDVIPMMYE
MTGIHKFFHMHVIFVFLFLVVGSGIAEKECITDDDCNRKYPMHANRGLQCLNGECKSSRIIKSR
MTKTLKFVYSIILFITLFLVAKNVDAKKCITFEDCPISKTRVYKCLHGECRYTIPYIPKVPKVK
NQIENYEVVKKIPLILPKQITGVFECPN SNCITHGEPVDSFVKVIAKKGNIHLKCKYCEKVYSQE
QRSGRGQRCPGEVFNQCGSACPRVCGRPQAQACTLQCVSGCFRRGYIRTQRGGCIPERQCHQR
NSSILMMHMTTRICEEINWCSNEFKFVTLSDTFSTGSSIMPQKNPDMAEILIRGKSGRVYGNLI
CCGGIKKEIEATKKEQEAIKKKIEAIEKEISGIVQQQNNLLRAIEAQHLLQLTVWGIKQLQARIL
DPRGKPCETVFORLSYNGRSSVVQCRPLTGRTHQIRVHLQFLGHPILNDPIYNSTAWGPSRGQGGH
DQYCPKSSITACKKMNIRNDCKDDCTGGSWCCATPCGNFCKYPADRPGGKRAAGGKCKTGYYV
DQYCPKSSITACKKMNIRNDCKDDCTGGSWCCATPCGNFCKYPTDRPGGKRAAGGKCKTGYYV
EGTWQHG YGVSSAYSNYHHGSKTHSATVVNNNTGRQKDTQRAGVWAKATVGRNLTEKASFYYNFW
GSLKSSQCNPECTRRCSTQYHKPCMFCCQKCCRKCLCVPPGFYGNKAVCPCYNNWKTQGGPKCP
GVSDGDRVTISYRSFAFGKALRGSSAQGFTFVLGKGQLEQFERQIVGMKKGESKEFKLEFPLDYM
KDSGSGINPQDMPKLFTKFAQSQPVATKNSGGSGGLGLAISKR FVNLM DGH IWDSEGPSKGCTVTF
KELSNRRGQILDMEQEGDMMTIKSKAPVSEMFGFAGAIRGATQGRCLWSIEFGGFEKVPTLQPI
MAEIVKYVYVIIIFLSTILVATNIEGTMSCFHDADCVHKRCQLPQIPKCVGKKCRCRGQYQANPMG
MFTMKKSLLLLFLGMSISLSLCKQERDANEERRDDPDENEENGGEAKVEEIKRAARPLGCKA AFC
MFTMKKSLLLLFLGMSISLSLCKQERDANEERRDDPDENEENGGEAKVEEIQRGARPLRCKAALC
MKRGKNMSKILKFIYATLVLYLFLVVTKASDDECKIDGDCPISWQKFHTYKCIQKCKWVLRFHEY
MQMEKNMTKTLKFIYVMILFSLFLVAESFFVDIMCKVHEDCPQKSTHKYKCVDDKCFLYYWEKGP
NISEVAPGGCIDVREVLMMGEGELTRIGSPYLEGVGVRAVVGECFRSRKITVYKYKSKDYHRTIG
PKSPSSAKSRLQTAPVMPMDLKNVSKIGSTENLKHQPGGKVQIINKKLDLSNVQSKCGSKDN IK
SYREAVLRAVDQFNERSAEANLYRLELDPPEQDAEDQGARKPVSVFRVKETVCPRTSQQPVEQCD
ATQGVQAQTVANFYLAYSMGLKLIPIINKIDLDAADIPRAMDQVESTFELDRENCLQVSAKTGLNVK
EQFIDVFRQNLHLSIKEPGLRFDVLQDPQVPTRFYIYEAYVDEQAVAFHKTTPHYKTCVEQLEPLM
LTEEAPTRAEMFTAMLESVGATPAEVPVVDDEGEERQVANTATEDLTTPGTYRATLTGVDAERALRF
MAEIVKFVYVMIIIFLSLFLVSIHINALNECTQDYDCPIEMCPFPFQPKCIMLKNLSIFSNSGIC SCT
MFTLKKSMLLLFLGTINLSLCEQERDAEEERRDDEKRDVEVEKRFVLGVVTKLLPSLFCMITKKC
MKKKREKKSEKKAILNLYQRNVRAEYEEERKRRLRPWLYDGT PQKHSRDPYWFDTVYNQMFKK WSEA
MQREKNMAKIFEFVYAMIIIFILLFLVEKNVVAYLKFECKTD DDCQKSLLKTYVWKCVKNECYFFAKK
MRTLCSLLLI CCLLSYTTAANSIIGVSEMERCHKKGGGYCYFYCFSSHKKIGSCFPEWPRCKNIK
QDFKTA AVDNRF EIIYQLLSMANNVRIRVKCKLKDAQIILVDSVSDLWPSANWVEREYDMFGIYFN
RTVLLKDLHDGKFIFYTNYESRKGTHLAKNPYISLSFVWHALERQVHIEGIAASKVPAGESD TYFRQR
SRSGRSGSGKGRGGRGSSGSRGSKGPSGSRGSSGSRGSKGSRGGRSGRGSTIAGNGNRNNGGTRTA
TAFTNWAQPA TEGPCPAQRCAALEASGEHRWLEGSCTLAVDGYLCQFGFEGACPALPLEVGQAGPAV
TPAPPAPPSGPGTGAGAPPTAAPPTGGASSNSAAPSPLSNSILPPALSSQGEFAATASPLQRMASI
VASGRTYHLKGVNPPKTEGVDDTEDEPLIQRDDDKETVKKRLDVYSQOTRPLVDYYSAWAANGDPAA
YLTQRDQLRGVNSLVDARHEDDETEDIQMYRWLAYEELPTLIVATKSDKIARGKWNQAVSQIKSLEL
AAAAGQSALEAAGLEATSVDLILLATSTPDDLFGSACQVQAALGATQAVAFDLTAACSGFLFALVTGA
AGLRTLILAYRELDEKEYKVFNERISEAKSSVSADRESLIEEVTEKIEKDLILLGATAVEDKLQNGVP
APLLAALGAADLALTTVNELVGNMRERAEEETRIDTRSRVEESRARVAKLQEVLP EHLSELREKFTADE
ARIKTRLFTIRQEMASRGLWDSFVSPREAKRQLKKDLKSQEP CAGLPTKVSHRTSPALKTLEPMWMDS
EEEPAPKPRKKRSRAAFSHAQVFELERRFNHQRYLSGPERADLAASLKTETQVKIWFQNRKYTKR
FAGVMNFSAYWYSDVLVKMYNAEPLPNNHFVNIIISELAHRAGTPVPKVYLINNSTPNAFATGRNPE
HVIQGDFTEEAPLQALLQALEGRPVDLVLSMAPNLSGVRVVDQPRAMYLAELALDFAGQVLQPGGDF
LPENAYEELKRQFGLFVRDVRGGNLLVVKTAEGHASGIAYLLDRLRDEIVGTLAGDDTILVVARTE
MKFTATFLMIAIFVLMVEPGECGWGSFFKKAHVKGHVGAALTHYLGDKQELNKRAVEDPNVIVFE
MQRVKNMAETLK FVYVLILFISIFLVVIGCD SIYYPISRCKTDKDCPNRKNYK GKCRKGFCMSSRLR
MTQYLVTTFKDDSTGQPH EHTTARDNQFTTVVEAESKEEAERKYE AQVKRGA VIKLGLFENIRECGK
CRKSCWVIRGHRKRCRSGERVKKPCSNGDYCCIAKKINTVPQAPKNAFSRNFRVHSRTAPVAVLKNST
EATAPIVGYSPIIGEKPLRGMADTCLSVIGVDSTAAAVGRHYGARCATGILDCWLVDHGDHAEIDGVTV
GFFGNTWKKIKGKADKIMLKA VKAVIMVKKKEGIFKEEAQAKVDAMSKKQIRLYLLKYGGKALQKASEKL
GGDGETREVRLGVGEQPLLPKPGESFEYTSGTATISSIAGSMKGSYQMAEDGFHFSVEIPEFILSVPR
GKLKELNGTAPEEKLTEDDLVLLEKILSLICNNSSEKPTAQQLQILWKAINWPEDIVFPALDILRLSI
HAVEYQFKIPHGHAVMIGIIYQFIVSNII LNTHYDISHYIQYKSLNYPLEVIQSFEFNSLYGLMLKDK
HGEGLVLVGAPLKGRVLIIDDDVISAGTSVRESIKLIEAEGATPAGVAIALDRMEKGTGELS AVQEVEKQ
MAEIVKFVYVMII FASPFLFSMNLDS ENICDGDYDCNPNEWCPPNYVLKCIN YQCSCIGFTPAIYALD
MAELRCMCIKTTSGIHPKNIQSLEVIGKGTCHNQVEVIATLKDGRKICLDPDAPRIKKIVQKKLAGDES
MFTMKKSLLLLFLGTINLSLCEEVNRNADEEERRDERDVEVEKRVIPFVASVAAEMMHVYCAASKKC
MNWKA FRLVSLYLLALVIAFGLIVVITLVFGASVTIAIGRILGNLIILKVIWDM LHK EYEEAKEDLEK

MQMKMATILKFVYLIILLIYPLLVVTEESHYMKFSICKDDTDCPTLFCVLPNVPKCIGSKCHCKLMVN
NSAVKDVEKKLNKVTKFIEQNEKEFTQLDLEDVQVREKLKATSKAKKLEKQLQKDEKVEELKSVPK
VKGNIKEPEVCPADNVRCIKSDPPQCHTDQDCQGIRKCCYLHCGFKCVIPVKELEEGNQDEDVSRPCP
AELRCMCIKTTSIHPKNIQSLEVIKGTNQCVEVIATLKDGRKICLDPDAPRIKKIVQKLAGDESAD
EAEEDGDLQCLCVKTTSQVRPRHITSLEVIKAGPHCPTAQLIATLKNRKCICLDLQAPLYKKIICKLLES
FAFPVIFGELYGYKDNLVGLMFIPVIGALWALATTFYCENKYLQIVKQRKPTPEDRLLGAKIGAPFAAI
FPNPHSFTGENVLELQGHGGPVILQWLLERVVQLGARLAEPGEFSKQAFLNKDLAQAEAIADLIDASS
FRVKKNADFKATFRVGLSVSKKLGNVAVTRNQIKRRIRHNFVKHSHLDFVVIARQPAKDMTTLEMEKNLL
FTPSVSFSQNGGVVEAAQRGYIYKKYPKGAKVPNKVKMLVNIRGKQTMRTCYLMSWTASSRTAKYYYYI
KARNAARIVVAFIVAGSPPTVLERHGLSEDAVNAVREALNNAFTKGDWGGVAKSVTDEMIDIFSISGTPD
LAGYTGIASGTAKKVDAIDKGAAAFVVISIISTVISAGALGAVSASADFIILTVKNYISRNKAQAVIW
MAEILKFICFMIIFLSSFIVSESLNGHGRNRCFRSDCPKVMCPSYLVTKCFKKHCRCKPGLQVQLNPK
MAKFVSIITLLFAALVLFAAFADPTMVKAQKLCERSSGTWSGVCNNNACKNQINLEGARHGSCNYVFP
MAKVLKLNVMIIFLALVLVAMNVNADVINCTQSDCQSIGLSHLKPKCTMLGFFFNFAVFGICECDQVM
MFTLKKSMLLLFFLGTISFSLCEEERGADKDDGGEITEEVKRGFLDVIKDTAQNLFATVLDKIKCKVTKC
NRLRRREFATAVRAGFVVSKAVGVAVVRNKVKRRLRHLMRDRIDLVLVVVRALPGAGDADHAQLARDLD
SVPVEKDVEGVNPNANLGLLLYDEEIFPPCTAEAAVRILERETNLSGKRVTVVGRSVTVGKPLALMLLKKG
VFTGGASLALAAVGLAVMADEIVKAATGVFSFIQALNPIMEHVLKPLMELIGKAITKALEGLVDKKT
EQILRYGRGIAPLWISGENTPKEKDIPDCPCGAKRLEFQVMPQLLNYLKADRLGRSVDWGVLAIFTCAES
FSKVWLQEVLRQCYFDGAIISDDMSMHAATEFGDAPTRVTAALKAGCDLVLVNDPVAADDEVLSQVTWET
HVDLPVTSRPFALWQGLQARYGHVSAERALSGNGLLALYEISCALDGVAVRASSAAEVGALAMAGDAQAD
KRLLTARQFSAVFRLGLVIGKKNVKLAVQRNRLKRLIRESFRHNQETLDIVVIARKGLGELENPELHQFG
MARILSFFYALLIFVSLFLVTTNGSLPDAPPCLFTPECPPDMCPTDLTKCINLSCQCTIEYDIDPDVVP
MDLTAEELTFLACLVENNNEEVQDSINIYKSFDKDAAAKVLEKQLLESIRLASRLREESKKSMLIGGLSIC
MVKTLKLVNYMIFFLSIFLVVKNVDGDDVVFQYVFDGCRIDADCPISGLQLLWMCINNECFNHVRPRYV
PKISYDAKIELATTSFEKILGDVQVVS DYLTIKAKPTGVEFAGKGDSEAAIDVKKDDDSMEALKVKKSE
PMIIFAKDGHFALEELAQAQYEVVGLDWTVPKKARECVGKTVTLQGNLDPCALYASEEEIGQLVKQMLDD
PNVGKSSLFNRLARERIAITSDFAGTTRDINKRKIALNGHEVELLDTGGMAKDALLSKEIKALNLKAAQMS
SLQPGAPNINNKDQWPQVSPHISRDDNGNTRTNINVQRHGENNDFEAGWSKVVRGPNKAKPTWHIGGTHRW
VDVIFSDVAQPDQARIVALNAQNFLRNGGHAVISIKANCIDSTAEPEAVFAGEVNLKKEEFKPLEQNTLE
YPDFSLHVKVAKPVAQVDARIKKIVADMADTMYEAPGVGLAATQVDIHERIVVIDVSDEQNELMVFINPEIV
AVEDGLIDGGGTATGRQEALLAWLTSLLDHGTIWNDAVLLAEAWRVWNPQSLTEVSDLALALAGSRERYQE
MAEILKFVCFMIIFLSSFIVSESLNGNRHGKDRCKFSDCPKYMCPSSLVAKCICKLCSCKPGLQIQNLNPK
MEKKS LAGLCFLFLVLFVAQEIIVTEAKTCENLADKYRGPCFSGCDTHCTTKEHAVSGRCRDDFRWCCTKRC
MKTLLTHHIAVQCTCGATGQTGDFLVDIEHFKLTGELIALDGMVFRSLTALYAYANTYGLPTKRLNTPIVYQS
MQIRKIMSGVLKFVYAILFLFLFLVAREVGGETIECETDGDGCPRSMIKMWNKNYRHKCIDGKCEWIKKL
MQRLNTEVVMILYVMIIFLCLLLVTMNVNAVICKFQSDCPKYMCMFPLKPKCVIILVFPWPWTAQCICD
MTIVQLLAVIAVLLLVPVTVLSFLRMLVWAAETWVRCE TGRGNRGTA AVATVLYGVATWVCAYLFFVVGAYL
NPLPVLDFAVATALRATEDLSQEQQAVVDLSDTLIEEVKAADTLVIAAPMYNFTIPTQLKNWIDLARAGVT
PAHDQNDYNTGKRHNLEFINILTDDGLLNEECGPEWQGMKRFDARKKVIEQLKEKNLYVGQEDNEMTIPTCS
RSKRDTCFDELTRLFGDGLSDDPKLEYEVYREFEEFN SKYNRRHATQQERLNRLVTFRSNYLEVKEQKGD
SIVPLEVIVRNVAAGSMARLGLLEEGTKLKTTFEFSYKDELGDPLINSYHAVAIGAATFEEIDTILEMTA
TQFLFLFIYSLIIFLSLFLGEAALERTRTMTLSYNIGCKSDADCPKAIIEPHYTRCVDGHCWLYFGEGPKLHN
AVNGITTFNEGYSISGAMVTAANKVIAVLDSKFNRRGFNQVLPKIDIIITDDAVSEVDKLALQKTRVKLI
ERELPSQFGVGNLTNLQLNDEDFSMQQIADTINRVRGYSATALDARAIQVRVPSGNSSQVRFLADIQNMV
ERLRKRPDFLLAARVGFTATKKIGGAVERNRAKRLREAARLVLPDYVFIARGGTGTREWARLLDDVKTALI
FKLLEIDEKYHILRGARRIDLGAAPGWSQIAAKVTGSTDEDIRVAIDFLEMAQLPGVKILQLDLPDSAP
GGSKCVSDTPGYCRTHCHRGETALFMCSPFRKCCISYSFLPQPDLPQLIGNHWPSRSRNTQRKNKKQQTTPV
GGSKCVSNTPGYCRTYCHQGETALFMCNASRKCCVSYSFLPKPDLPLQLIGNHWQSRNRNTQRKDKKQQTTPV
IYYALLKSDECLLITINSAGCVIETIYIAYLVYAPKKAKMFTAKLLLLNVGVFGLILLLTLLSAGDRRIV
KSDVWKISGHYDNYKENMYFTTIDEQEYGIKPMNCVGHKIVYQSALHSYRDLPLRFYEGVVRHEKSGVLHG
LLLQQLLPVQFQEVDTDFDSDPDDMEEFEEYLEEFHRTGTPRPTKEKVERRVLEIEPGMPLYDRNYCNIEIM
MAKEIINNTERFILVQIDKEGTERVVYQDFTGSFTTSEMVNHAQDFKSEENAKKIAETLNLLYQLTNKKQVRK
MFTSKKSMLLLFFLGTISLSLQCEERDADEDEVIEEVKRGFLSVVKGVLKGVGKNVAGSLDQLKCKISGGC
MQKRENMTIVVKFVCMIIIFLSL CVFSMH IETVTTCTIYSDCPEDMCYPPKKSFCSTFEILSIERKVGVCCEI
MQRSRNMTTIFKFAYIMIICVFLNIAAQEIENGIHPCKKNEDCNHMCVMPGLPWCHENNLFCYENAYGNTR
PRRHGKTWFLVPLIALALATFKGIKIGYTAHIRKATEPVFDEIGARLRQWFGNSPVDHVKGENISFSFPDGSK
QDWYPVLAKAEGTDAERAVALKQLREEILAIPIFTQMPYFMSEEFSLVDCYIAPLLWRMQLGVDFSGAGS
ADPSRIDVVHPGVLDLTFTPGDQAAARAALGLDPRETVVAFVGRIPQLKAPDILLRAAKLPDVRVLVAGGPSG
AGDPLADPN SQIVRQIMSNAAGWPPLVPERFRGMAVGAAGGVTTQTVLQGAHAHMPVNVPIPKVPMGPSWNGSKG
ERLRLRDFLLIFRLGIVVKRFGKATRRNKLKRWVREIFRRNKGVIDIVVI PRKKLSEEFERVDFTVREKLL
FDGDDVYTGDMYSADKQTVVYVYVNLHGLISGLYGVLPKPLDMIQPYRLEMGTCLKNEQGDNLVEFWGHEITDVIN
IKSFSAWQREQMNDTLAYLDWPVLARLYYACKADDADAFARWSRFLANRETAELRLEEQQRGAALARLLDGWQ
MFTLKKSLLLLFFLATINLSLCEEERNAEDDGEMTEEEKRGFMSTFKQVGISAIGKAAKNVLDVLSCKIAKTC
MFTMKKSLLLVIFFLGTISLSLCEEERNAEDDGEMTEEEKRGLMSTFKRVGISAIGKAAKNVLDVLSCKIAKTC
MFTMKKSLLLVFLFLGTISLFFCEEERNAEDDGEMTEEEKRGILSTLKQFGKAAVKGVAQSFLNTASCKLAKTC
MIETLKFFVYAMILFLSLFLITSEVGGLYIGCETDRDYPLANKTFYLCIDKKCEWTVTDSLSTRSGRMQKLSI
MQKDKSMVQIVKFVYVMIIVLSSFFVVAINS DGYLECTTDYDCREEWLCPDMEAKCFVSFALARFLSKGKCLCV
RFAHRVLSDAEFAVWKARSERWPKRGLSYLATRFSAKEAFSKAIGLGMMPMSWRLLCEIANLRSGKPVIVLHGE

RIGAWNLARFAEALLPLADDQDEAIKLAEEISNYAEQFEHHLNMGRAKLGLFNEEEDLSLIKDLLEVMMHK
SDDPKESEGLDHCVCVKTTSLVRPRHITNLELIKAGGHCPTANLIATKKNRKLCLDLQAALYKKKIKKLLES
SRNLIRDGLVDLIRTHYPACECIAGVATAGIPQGAIIADSMNLPMIYVRPKPKDHGMQNLIIEGKITKGQKIVVV
TGPGLHIYKSGFLDGLSDDAKRAAIAKIEAAGQGEKGVNYRLRDWGVSRQRYWGCPIPVVHCEDCGVVGPAAD
ARLKKCFNNVTGYCRKKCKVGERYEIGCLSGKLCCINYEKEEHVSFKKPHQHSGEKLSVQQDYIILPTVTIFTV
ISFVLAWFEEGEETITAFVEPVFILLILIANAVVGWQERNAENAIEALKEYEPEMGKVYRADRKAVQRIKARDL
MFTLKKSLLLFFLGTISLSLCEEERGADEDDGVELTEEEVKRGILSTFKGLAKGVAKDLAKGLLDKFKCKITGC
MVEAAGDKTKDGASWLGDKIGDVWDYVQHPGKLVNKVMSGLNINFGGGANATVKIAKGAYSLLKKEISRQSKIVV
PDEVLEIAARAKAAGADRFCMGAAWRSVRDGREFDSVIEMVRGVRELGLEACVTLGMLNAAQAGRLKEAGLTAYN
TVVDCGILLPVDSDARNLILRRLGSRTANFAIESAMTIEQVEQALRNGQIAAEKIAQAGADLLMFGEMGIANTSS
DAIIDGLAAEEKQIESGVFLPMKFLSVESPETFNKIHNHGTWAVAAVGMTGYVLGNDELVEISLMGLDKTGKAG
GWMSEKKVQGILDKKLPEGIIRNAAKAIVHKMAKNQFGCFANVDVKGDKRHKCAEDKEGICHGKCKCGVPISYL
LYLWMVRWRIAIVAAKIKTKNQPLFRILVLGLATNNITPAKSGGEALRAYLLSKYSKTPFESSFATVLIDKGLDTF
MASAATFQSSYDSPVKACIPEARPATYETSVSSSFIPPLQIKRVPKSGRPEGSHVRRPSRESRNMLELDLDFD
MFTLKKSMLLLFLGTISLSLCEEERGADEDDGEKEVKRGIFSLIKTAAKFVGKNLLKQAGKAGMEHLACKANNQC
MQKARNMAKLVKLIVYVIVFYTLFLVATEIVSGIPCNDVDCPQTLCEQLIADFKYMIDFKSECVSRMCACTGSPV
MQRMEHTTKIQFCVYVLIIFLSLFLVVTNGDKPRYTTPRNAVKIAECVSYTDCQGGCPACYMRCIDGQCEPFIKFI
MSLLALVAGTLGVSQSIATTVVSIIVLTGSTLISIIILGITAILSGGVDAILEIGWSAFVATVKKIVAERGKAAAIW
IPVLCPVCTSLVGKLDLVLGGAVDKVTDYLETLCAKADGLVETLCTKIVSYGIDKLEKILEGGSAKLICGLIHAC
MIKQFSVCYIQMRNMTTILKFPYIMVICLLLLHVAAYEDFEKEIFDCKKDGDCHMCVTPGIPKCTGYVCFCFENL
SKCKCSRKGPKIRYSDVKLEMPKYPHCEEKMMVIITTSVSRYRGQEHCLHPKLQSTKRFIKWYNAWNEKRRVYEE
SNSSNAEKKLYELDVDRILRGEDSRTTLMIKNIPNKYTSKMLLAAIDECYCKGTDFLYLPIDFKNKNVGYAFINLI
SPRPDDKKNQGSASVDVQNERGEGTKVDARVRQELWRSDDGRTRAQAYGHWDRTYGGRNHGERSYGGGMRIEHTWGN
SVIGCEICEWLVAEAGFVNKTQPIEQELLQICAKLGPYEQICDQLVLMELPDIIIDQIIAKEPPAIVCSQVKICNG
VPLSRTVRCTCISISNQPVNPRSLEKLEIIPASQFCPRVEIIATMKKKGEKRCNLNPESKAIKNLLKAVSKERSKRSP
FEHSIKNVDKGNVVAIVGTQWGDGKGIIDILSKYSIDITCRFNGGGNAGHTICVGDKKHALHLLPCGVLYENNINIL
LTTIEAERTELLRLRIENFTTLRRRAFMESEFAIDRNFEIIFAHLSDDGSLQLDNPEDPFSSGLNLIAHPKGKPVRL
MQMRQNMATILNFVFIILFISLLLVTGKYREPSSFTGPTCKEDIDCPSISCVNPQVPKIMFECHCKYIPTTLK
YIGFVEEKGALYCELCYEKFFAPECGRCQRKILGEVISALKQTDWHVSCFVCVACGKPIRNNVFHLEDGEPYCETDYA
HHSSAFTHVIRDDGGVETYLDSITEAEQELLASAKQYDLFDDDDSDDEEATLRYKLKDRLRKRRARLQAWPPVKY
KVSRLTIHGGVGAITESDVMLAAASNAIITGFNVRTPTATAISKKEKVDIRTYRIIYNAIEDIEAMVGMGLDPEFKEV
LQLEMLRDNILFRLGMASTIPAAARQLVNRHILVNGRIVDRPSYRCKPRDIIMTKDNQKSRTLQNSLGSPPQKELPN
LTVPCENPTTCEPDEFCTPPMITRCINFICLDGPEYAEPEYDGPVEEYDHKGDFLSVKPKVINENMMMRERHMIKEIEV
MAPTVMMASATAVAPFLGLKSTASLPVARRSSRSLGNVSNNGRIRCMQVWPYGNKKFETLSYLPPLSTGGRIRCMQAM
MPVAVGPYQSQSPSCFDRVKMGFMGFAVGMAAGALFGTFSCLRFGRGRELMMGGVGKTMMSGGTFTGTFMAIGMIRC
PTPVRTTQNVGNQNDIMQGGMLIQFNPSYDMSADHSFSSSNMLPATPQKFDTSFTTSQIHNVSRSPLSPKAGTSPRDI
QVYGASAVPLIAQTSVAADNSIIKKGSVILLKIPILDQNGVFLNKYEMRLVIALDVGGAIKGHHFDIYEGIGTKASILS
VACVERDDIATLVKLPVGKKTAAERLLVEMRDKLKSLLETSSAGSEREFMLKSNYTPAPVVNTAEEDAIAALLSLGYKPA
VHTAPGHGEDDYIVGQKYDLPVISPLNDKGVFTEEGGPFEGMFYDKANKAVTDLLKEKDALLKLDFITHSYPHDWRTKK
AIVLVLLALPSLQILYMMDEIDEPDLTLKAIGHQWYTYEYTDKDLSDFSYMTPTTDLPLGHFRLLEVDRHIVIPMES
EVIDEVMLENEPGYWKYYRPTWSQAMVDIHFSLSDRIRYYPHPRIRQSVEKLIANLNNVTLPPLGLISQFMPVQFERLSE
ISLTTILILSVAVLLSTAAPPSCRECYQSLHYRGEMQYFTYHTHIERSCYGNLIEECVESGKSYKVKNLGVCGRNGA
KALKNCTVNEPFFQGHFIKEPVFPGVLMIEAMSQAAAVLIFKISIGKLNINQLYFVGIENTRFFKIVVPGDQIFIEVIYL
MAKVASIVALLFPALVIFAFAEAPTMEVAQKLCERPSGTWSGVCNNNACKNQICIRLEKARHGSCNYVFAHKCICYFPC
MFTLKKSLLLFFLGTINLSLCEQERDADEEERRDDDEMDVEVEKRFPLAVSLAANFLPKLFCKITKNVETLEMELEII
MKTFSVAVAVAVLAFITQEQSSALPVGTGIEELVEPSSDNNNDHQLVVELRERLVNIRKKRAPTDICPYCTGDGFH
MPFHYFPLNPTIWPFGWIPGLGKDRCLLPKVTGPCKASLTRYYYDKDTKACVEFIYGGCRGNRNNFKRDECEKACTDH
AGANDLCQCECEDIVHLLTKMTKEDAFQDITRKFLQECDILPLKLLVPRCRQVLDVYLPVLDYFQGGIKPKAICSHVGLC
DFGVYAYTTAYIGANFFSPVENADIRQMLTFAALAIAFLLRPIGGVVFIIIGDKYGRKVLTSTIILMAFSTLTIGLLPSY
ERTFEHSYRVTNQLTTSSEVDIETFGTPEAPEKYLVALGCATWTSGLKEKEIADNDWLVPANNHILFDVPWAEWCWAAQ
HQAIFIQRPHSDAIAYQAAAGIIGQFSQPPIEPHQFVLMYRSDHLSPEERQLPPTFLYAMD LGNDVYFVEETSLAACPAI
LPRGWEKLVKDFVDWLPKRLDEYTKAALQNSILKGRVTGVAAYNTKEALEWGTGAGLRSTGCDLRLKARPYSGYENFEF
MAVMDLSGIVQSLMKTSLSSRFWKSQKKASLKALESKIRELNHQQELFQKRDRINKQAIQLGSKARRLEEAKALIEKYI
MKLFYVLIYFISLFLVINVQAFFDCENHDDCKNKIKYVLPRIAECRDYKCNCFPLNLSKTLWSASTKRVHKSQAQTNDFLH
MLTEGSYSQQKGNPLCNNQIAGVLKKTTKALNMNKVTTTHTFRHTHITLLVEMNVSLKAIMKRVGHVDEKNNHSHIYSCN
MRLVVCLVFLASFALVCQGGVYKGGYTRPIPRPPFVRPVPGGPIGPYNGCPVSCRGISFSQARSCCSRLGRCCHVKGKYS
TEDVQLEFADDGIRRLAEIAFAVNEKTENIGARRLYTVIEKLLLEEVFSFAGNHAGERVTIDAKYVDRLAGEVSQDELSRY
TTGMTGAISYHPYGSAAYPYQLNDPAYRKNA TRDATATLKAWLNEHRKNPYPTKGEKIMLAIITKMTLTQVSTWFANARR
VSDMLEIFLIGFVYLFYRIAIIFLKSCKTIHTNIYEMLMLATIFMISTIAKHKQTHILIAFLVMFFMSKLLKQVQGSYEE
GGYGNMRLVHRLLPNQLPDDVSITLIDRNPYHCLKTEYYALAAGTISDHIRVSFPEHPRLDVQYGDITSIDIVQKQVLFQD
IIVDGGDGTSVLNSLTGDVCLDGLTFGGFNFPVNPSELKSKYIIFYSSKPNIFKITRALDRHFNDERRDIIINVISNLTRV
MAILEDIFEELKLLNNNLRVLNTELDSTVDSSIVQEKVKEAPMKEETAQLESIEEVKETSADLT KDYVLSVGEFLKKSRYF
MALLNKLLCFALVFMIFGEFVNTPD CYEDWSRCTPGTSFLTGLWKDCHSRCKELGHRGGRCVDSPSKHCPGVKLNKQCHCY
MPQLCEAIISPALTGOALEVGDRAFEQGNTPAANDRVATLEGQVQALLALLQGQQAAGAPVAPVEVVAEEVKAPAKKATAEK
MRLVVCLVFLASFALVCQGGVYKGGYTRPVPRPPFVRPLPGGPIGPYNGCPVSCRGISFSQARSCCSRLGRCCHVKGKYS
MSMKAKYFQMKRKS KSKGEIFYGDIVSDKWFE SDVTATDFKNKLDLGDISEIDVHINSSGGSVFEGHAIYNMLKNASCKN
ATPGGYRKAMRLMEHADRFRPLILTFIDTPGAYAGLQAEEQGGEAIAVNLRDMFGLRVPVIATVIGEGSGGALGIGVADRL
GRDYRTCLTIVQKLKMMVDKPTQRSVSNAATRVCRTGRSRWRDVCNFMRRYQSRVTQGLVAGETAQQICEDLRLCIPSTGPL
LPRTFLLCLHPLYVYVLLLEILLTLLRILMTIILGCDLEPHFHEPYLATSLQDFWGRRWNLIVSASLRAIVYTPVRRVCQVRMS

MADKGVCSRLSALFLLVLLVISIGMMQLELAARTCKTPSGKFKGVCASSNNCKNVCQTEGFPSPGSCDFHVANRKCYSCKPCP
MKFFITFTFVLSLVLTVYSAPREFAEPEEQDEGHFRVFRFTCDVLSVEAKGVKLNHAACGIHCLFRRRTGGYCNKKRVCICR
MRLLLMLPPLLALLPQVIPDYSAEKRCLNRLGHCKRKCKAGEMVMETCKYFQVCCVLDDNDYKQKASITRTMEKTSTIEYNLS
NLAKGKEESLSDLYAELRCMCIKTTSGIHPKNIQSLEVIGKGTCHNQVEVIATLKDGRKICLDPDAPRIKKIVQKKLAGDES
PGDIVLPKPRYSGFFNTPLDSMLRARNIRHLVFTGIATNVCVESTLRDGGFFLEYFGVLEDAHQAGPPFAQQAALFNIETFF
AAFSESSKGLLATSAAKAPGLIPKLDATYDVAYKAAEATPEAKYDAFVTALTEALRVIAGALEVHAVKPATEEVLAAKIPTGEL
APAEEGVVECNKSLDPTTTTFLKMAADSGGLPQVTQPGKLEAFKYFLQGMGYIAHLKDRRLSGGAVPSASMTLARSRTASLTS
CTSCRKARAWLVKHEVDFOEHNIITSPLSRDELMSILSFTENGTEIISTRKVFQKLDIDVEELSISGLIDLIAKNPSLLRRP
EEFSKLRRLSYPQTDVFLLCFSLINPSSFSNILDWVQELNENCNPTPIVLVGTQMDLKSNSVILDRCEKKQLPISTQQGIEL
FRKLPVNRPIDWKVCQIRIVGLLGFAAPFTQCGYPALMPLYACIQAKQAFSTFPTYKAFLSKQYMNLYPVARQRPGLCQVFADAT
GFQHQYTAYIAISYFDLFLHKRFIQLQKDETWAMRLLSVACLSLAAKMEERIVPGLSQYPQDHDVFKPDVIRKTELLILSTLDW
ILTRAALKASGWERGRVIGSGTVLDTARFRYTLKECGVDARNIHGYILGEHGDSEFAAWSMTSVAGRRIDEYCSGGVCSGPH
KFAK
LCPLVSFCLLLFFIDYLPKMLVKKIGIVSIFISMIITFYSLDFLNCGKQCVFYIPLWVWISIDYLIKIDFNFMLDGLSITMLTM
LILMDIRHPMTDFDVLMLDWAIASNMPMHILLTKADKLTYGAAKNTLLKVQAEIRKGWGDVSIQLFSAPKRMGLEEAYITVLA
MKTCSVAVTVAVVLVFCIQOSSASFPEVQELLEEAVSNDNAAAEHQETPVDSWMMPNNRQKRGFKCKFCCGCCRAGVCGLCCKF
MKTFSVAVTVAVVLVFCIQOSSATFPEVQELLEEAVSNDNAAAEHQETSLDSWMMPYNRHKRSFKCKFCCGCCRAGVCGLCCKF
MKTFSVAVTVAVVLVFCIQOSSATFPEVQELGEAVSNDNAAAEHQETSVDSWMMPYNRPKRSFKCKFCCGCCRAGVCGLCCKF
MKTFSVAVTVAVVLVFCIQOSSGTFFPEVQELLEEAVSNDNAAAEHQETSVDSWMMPYNRQKRAFCKFCCGCCRAGVCGLCCKF
MKTFFVAVTVAVVLVFCIQOSSATFPEVQELLEEAVSNDNAAAEHQETSVDSWMMPYNRQKRSFKCKFCCGCCRAGVCGLCCKF
MSDINIVGMSFYLTDDTKRFTDFPTEKGVAGWNLVYEASNTGGFVHRLVRNSVTASAEILLKNYDSKTSSGPWTLHEGRIIS
MTIIDYGSKPLNNCSKNELLDIQSQLENEVKLIKLIERTTKEVEEKEVPHYKNYISPEKRSYLEGMERVENIVRNFYGFNNE
NKDCKYWKDNLGLNYCCGQPGVTPPFTKKHLGRCPAVRDTCTGVRTQLPTYCPHDGACQFRSKCCYDTCLKHHVCKTAEYPY
SARVRTRAQYITVFDTARRTSDPLLSLHWRTGDTPPRLGMAVSRKVDTRAVGRNRIKRVLRDAMRHLLPELAGGDYVIVARSAA
TRRAGLTAEHGVRWESEGAGDYTLETVEKPQRGTEIVLHLREGEDEFLSDWKIKSVIRTYSDHITLPIVMKKTEWKDGVETPTD
DGSFLDYRTLALHRLVDHVLATGFTHIELLPITEHPFDGSWGYYQTTFYFAPTTFRFGAPDDFRYFVDHCHRHGIGVILDWVPAHFPR
EHRKVLASVAIISTQAPLSSLSQTSTAHEFVHRGNALQIYSSPNQGPHEQTLKRNLPKPTTELLDQTDLKQSSNEQGIAAIIITP
GSIEIFSRPKDHWPSSELLAKHMGILPQHSTLSFSFLAHEVAELGAMPLAISNHQAQQLAAKNMVKVGVDHLANRLYPTLSGGEKQ
KFINTMSQKPIMEKLVPLPELDNDSLGNKSDNWEYIYEPNPQTLDSLLVRYLESQVYQAVVENLASEQAARMVAMKAATDNAGN
LYVCRVSDCPENFCVPLTIQCINYTICDDPPYGEPEYDNNDDFVTLNREKAKIKNEEMMRERDMMIEIETYSVADDLPHL
MQNAQSEKVCENSAKLLTHLKSPEKTELKIDVSDRAADAIDIMRQNAEALARLQEMIAGKQLSAKDVAERSMDFGEVIEGELA
RFFLKDQIPTWFVAVGGYITIAATSTAILPHMFHQLRWYIYILVIYICAPVLAFCNAYGAGLTDWLASTYGLAIFTIGAWAGSEH
SLYREVMLENYSNLASLGFQASIPPVIGKLQKGQDPCMERAPEDTCLDFEIWPEIEALPPKQDVLTKETSHGLIKNGSTKCVYW
VGQITDYDKLTLEVWTDGSIRPDEAVSLAARIMVEHLQLFVNLNERATGVEIMVEKEEADAKEKLAEMPIEDLDSVRSYNCLKRA
YFQKDRGEFFAKSVKFKYPRQRKTVVADGIGQGYKEVQEISPNLRYVIDELDQICQDRDTEVDLKRKILDDLRLHLESVVANKINE
CLIGLYSMSTNLFHAERQFVVCVSETSERDLKLFANLNLAIYLRTRKRDTLKQILDVAVSTENTHTYSSQALMGFFYYVQGLHAF
DFGLDCDEHSTESRCCRYPLTVDFEAFGWDWIIAPKRYKANYCSGECEVFVLQKYPHTHLVHQANPRGSAGPCCTPTKMSPINMLY
DHHHDHGHDDHEHELTLEKIKEKIDYADKTPVDQLTERVQAGRDYLLGKGARPSHLPARVDRHLSKLTAAEKQELADYLLTFLH
LAQRIANRDVPSLTLEDKLVITLDVGLLVAGTKYRGEFEERLKRIMDEIKSADNVILVIDEVHTLIGAGAAEGIDAANLLKPALAR
MFTLKKSLLLFFLTINLSLCOEEERNADEEERRDERDIEVEKRVIPFVASVAAEMMQHVYCAASKMMLKNWSSDVENHLAKC
MKAFSVAVVLVIACMFILESTAVPFSEVRTEEVSFSDSPVGEHQPPGGESMHLPEPFRFRQIHLSLCGLCCNCCNIGCGFCKCF
MKHLNFFIKALKKEFESFKDTGASIEEMTKSKPYTKVGSQERAVLIEWVGMNRKNIHILNEHGYTRDGKKIYTKRFSYCKNISC
MKQFSVAVVLVMACMFIVESTAVPFSEVRTEEVSFSDSPVGEHQPPGGESMHLPEPFRFRQIHLSLCGLCCNCCNIGCGFCKCF
MLKEAVSESLWNAAYEILHTTMTADKVIALVTVYNASNLEESTYAYEAIQDSVNKAICGGAVEDVKWKQFVGSLSDSYKEFFEKV
MNFKLLNLFAMVLFGVVVISNAAKPPSGFLTTPDADNDENGNGVEEQTLERFTCDIWQNAACAIIHICANGFRGGYCAQKVCVCR
SESRLQSAAASSASKKKKKAKQAPASTGAKDRSVDRGLFCTHHEVMVAGLAMLGMDAINNKDQPVSVLLVGLGGGLPVQVRD
AEGATLSAADRADSSKLTIGADAGELVGLGKIGVTLAGDLAAMVDGDFVLIDFTHPSVTLKNLEVCRRQAGKAMVIGTTGFTVEEQQL
GKVTVERTETAEFIYCCVIAKKGDRREALVKICGGHTLIAEKRLNGESVFSVDSTQAKATGSICEGVDITIESIYRFAQEVFPFEEIKF
IGRSEGAEVHFNIDGTDEKFTVFTTRPDTLFGASYCVLAPEHALVADITTADQKEAVEAYINSVKMKSDLERTELAKEKTGVFTGAY
KPKYQPFAGKGQRLGSPTPGIRAPAPSEPAPAPQSSSGPPKPNVDESQPVVTLQIRLGDGTRLTSRFTNTHTIGDVYDFVSAASPQS
KVFGPAETQAKGATEIVNMGILQTEHYHIHLVLTAGHTEQHVSYLVDGHLFCGDSLSFAGCGRVFTGNVQHMFDGLQRLKSLPDDTI
MKLVVVQFFIISLLLTSSFSVLSSADSSCGGKCNVRCISKAGQHEECLKYCNICCCQKNCVPSGTFGHKDECPCYRDMKNKSGGSKCP
MLLLIITILVVVLAVVIYKSEKETVRECFGILVEGVTLAKAKSDILKVAKVWFYTLKLTGELILVPFLTIALILFIIAGATYRAIK
MYVVIVYDVGVERVNVKRSFLREYMNWVQNSVFEGELTAEFLKIKSRLKELIQESSDHIIFYSSRDRKYLGIEDLGTAKADTSNII
SLKKEKSNESIVAKSPVIAAATKGGSPSVQEQKIHEGVITESLPNGMFVWKLDCEDLVLGYSISGRIRRSFIRVLPGRVKIEVSR
TISSKDEDFLDLSVDVEQNTSITHCLRGFSNTETLCSEYKYYCEECSKQEAHKRMVRVKLPMILALHLKRFKMEQLQRYTKLSYR
ADKKNPLEECFREDDYEEFLEIAKNGLKKTSNPKHIVYPVKPSEQLYEESLRDQLPTSMHRYPSMIQKIFFAGEYTANAHGWIDSTIK
DKHMIIEGRKSCCRSTLGRNCYNLCRVGAQKLKAGVCRCKLTSSGKCPGTGPKMIEGRETSSATETTTETATKSEEAKETATEHDEL
DPSVRPYPTSGYPILGFTNLIFSQCYANATQTQVVRDFFTRHYGATANNDTAITNHRFVPLPASWKLAVRQSFLTSTNNLYIGHSNVC
GSPMHRVSLGDHWSWQVHPDIDSERHSPSFSVERLTNILDGGLPNTVLRKVESIIQSDPVFNLKLYFMTREELYEDAIQKRFHLEK
KVRHFGSGIPITETGKMGSKLVGIVTSRDIDLSEKDNKKYLEEAMTKREDLVVAPAGVTLEKANDILQRSKKGKRLPIVNDKDELVAI
LFANLDFNPIEDAVQQADMEVIDRYALVLARRLQERLAGGYPRYAFHFAVKDIVSFCEDLGAFYLDILKDLRYTTKDALRRARS
LLTGWHLLSVFPGDLVACLFLQLRLDLRLGYCPELSLCLDCGEPYIADSGWPAYSPMGGLVGNCCRHRHSEMGLIAPGALALLRLH
MHSDLGKIIQSLLDEFWKNPPVLAPTSTAFPYLYSNPSGMSPYASQGFPLPPYPPQEANRSITSLSVADTVSSSTTSHTTAKPAAPS
MKLKEFIKLAESKGATLEAYNELGGYELTRGDIVDQNPVLIAYMQGTHSVEISNEELENKELTELAFFVYKNASFYIPNEKELLSGGL
MKRKKMKLYSCFCIKYNPSMKDREILKATESKSLTIIMRSSKIEYLPQTNHLVKIDRGLYSKLFNIKEIRIDTPDIGYNTVVLSK
MKTFSVAVAVAVVLTFICLQESSAVPVTEVPELEEEISNDNAAASYEETSVETWMPFDIRQKPHSLIKCSYCCDCCVLGVCGMCCQ
MKTVLAFLFLTFIAFTHAESYEDVKEEIKNEVEREIFEDLEESDVLESNVRELNDAKPWRFRRAIRRVWRKVAPYIPFVVRTVGKK
MPYLIVTYDIAEERVNVKVRKILKKYFMWVQNSVFEGEITEGKLLKCKLELEKVIDKEVDSVYFYSLENRLNYRKTVLGIEKEITGNIL

MRKFVTTLTASPRNKKVGNHRLAISPVSLRRYYYFNTAICIENPVTREFAIDDSYGLSTNQCAQYRQYFSLGGYKEVSLEEIHAV
MWPKDENEFEQRSSLECAQKAAELRASIKEKVELIRLKKLLHERNASLVMTKAQLTEVQEAYETLLQKNQGILSAAHEALLKQVNELRA
VKEDIEKAGVCPADNVRCFKSDPPQCHTDQDCLGERKCCYLHCGFKCVIPVKELEEGNKDEDVSRPYPEPGWEAKCPGSSSTRCPQK
VKGIEKAGVCPADNVRCFKSDPPQCHTDQDCLGARKCCYLHCGFKCVIPVKKLEEGNKDEDVSGPCPEPGWEAKSPGSSSTGCPQK
VKGIEKAGVCPADNVRCFKSNPPQCHTDQDCLGERKCCYLHCGFKCVIPVKKLEEGNKDEDVSGPHPEPGWEAKSPGSSSTGCPQI
ALLAQLIQDKLDAYKADDPMTGEGPDKARSQLLILDRGDPSSPVLHELTFQAMSYDLLPIENDVYKYETSGIGEARVKEVLLDEDDDL
CHAITDALFGAAALGDIGRHPDTDLEFGKADSRVLLREAVRRVREAGFDIVNVDSTVIAQAPRLAPYVEAMRSHLAADLGLSPDRVNV
DMNLPERFKLAFTNERNHAEQPMIHRAILGSFERFIAILSEHFGGNFPFFVAPTQIALIPINEEHHVFALKLKEVLKKRDFIVEVLDK
EHDKYNIRSHGYKPHFVPNFNGPNAIRQFQPNQINENWKRHSYAGEQPETVPYLLNRLNRNTNPPGNWKKPSDSL SVASSSREGYVS
ETAVESVRYAKKYFPIVQWSAEDACRSDLPLATIVEKVEIAGANVINIPDVTGYITPKQYGDIFSFLKKNVRNIEKISLSAHCHDDL
LSPALKDLKEMFLDGKIIIFNNNPLMKWYINNVLKLDNRGNWLPSKQSRYRKIDGFAAFLNTYTDIMNKVSDSGEGNIEFISIKDIMR
MDNSKRIIKKIIFITISALVMVTL SKLFSKYVIVEQNAPFQALIGGASCALLSSILFDWYTNKKKKENVENQLKEAISDLQKIKAIKR
MFEEREKLFEFSERVSGARMHANYVRPGGVAWDLPIGLMDDIYDWAIFPERIDELEDMLTENRIWKARTIDIGLVSAAALNWGFSGV
MKFTATFLLLFIFVLMVDLGEGRKKKSKRKGSKGKSGKGRWLERIGKAGGIIIGGALDHLGGQVQGPDYDYQEGEELNKRAVDE
MQKKGYMAKLVTFVYVMYIFLSLFLVTKGAYYECSDNSACQATTKCVLPVPRCIKYKCLCGNSNGSGNRWSTRPNRIQKGSTESNYF
SINEPFFNGHFPHPIMPVGLIIEAMAQACGILGFKTVNKL PADGYVYYLVGSDNVRFKRPVMPGDQLRLEANVIRGKRGIWKFACRAT
SKMVRLTREQAEGFYAEHQGEFFVPLVEYMMSSPIVVSVLEKENAVKDYRTLIGTTNPETA AEGTIRKDFALSQRENSVHGSDSIESA
AISCQGVSSALSPCISYARGSGSSPPAACCSGVRSLAGAARSTADKQAACKCIKSAAGGLNAGKAAGIPSKCGVSIPIYAISSSVDCSKI
R
FPICAFSHCRDVVAAVSNAGGFGILGAVAHSPKRLESEL TWIEEHTGGKPYGVDVLLPPKYIGAEQGGIDAQQARELIPEGHRTFVDDL
L
MIQLSERQQDLLQVAEKYEQCHIEFYTAQSRLFGTEIMGEVVKTSGLTKIAHPEEDLFEVALAYLASKKDILTAQERKDVLFYIQNNL
C
MLNIFYNVWGGFISHPVLITLSIVGIYVLGYALVKFIVGIVVLSNSLNTKWVIASAIFTTVLSIVFTIYSFLVEIYVVLVVISGLVLL
T
MSAVPFTRVLLISGFLAHL LLLSTFVTLTVCKEVTEESDDL SKRNVLQRQLWAVGSFMGKKSLENTNRRSDEDMEISALFRGSPLKVKRS
D
MSAVPFTRVLLISGFLAHL LLLSTFVTLTVCKEVTEESDDL SKRNVLQRQLWAVGSLMGKKSLENTNRRSDEDMEISALFRGSPLKVKRS
D
NNFDLVNFTSESIFNFMRDASAKAKGRNYLWYFFHAPMYNIVRAEEAEIEILQSSKLITKNMIYELLKPFLGEGLLISTDQKWH SRRKAL
T
AAAQREGRTCAFIDA EHALDPVYAKKLGV DIDNLLCSQPD TGEQALEICDALSRSGAVDVIIIVDSVAALTPKAEIEGEIGDSHMG LAAR
MM
AITCGQVTSNLAPCLAYLRNTGPLGRCCGGVKALVNSARTTEDRQIACTCLKSAAGAI SGINLGKAAGLPSTCGVNIPYKISPSTDCSK
VQ
FGVVCREHPDGL EIDGIDRSTLRHPAGGVFCYDDHRVAFSFSILSLVAPTPTLILEKECVGKTWPTYWDALKQKFGVSLKGKELAESEI
TH
GGELFDKITKRGHYSEKAAAEIIRS VVKVQICHFMGVIHRDLKPENFLLSSKDEASSMLKATDFGVSVFIEEGKVYEDIVGSAYYVAP
EV
LNCGQVDSKMKPCLTYVQGGPGPSGECCNGVRDLHNQAQSSGDRQTV CNCLKGIARGIHNLNLNNAASIPSKCNVNVPYTISPDI DCSR
IY
LVLVAQDTSAYGHDISPQSALPTLLRELSKLDGLEWIRL MYLHPLYLSDDIIDVVAYENKVL PYLDIPIQHASSKILKLMHRRHDNSHL
RT
MTKLSSAWLVKFEYRAVKMRAFTSYDKAVTYYNEMVQNIDEQIEESYETKTTYTDDHKPLL TQFIYWNEDRTFRN ILGFVSIKEIELE
EN
MYVVVAYDITEDEV RNKVADALKAYGLERIQRSVFVGRINPALLKDLVERLKRITKGANADITIFKVD RRAIDTAIRIGPPPPARKNVD
LY
RASHGNSITHRSHGSTGQRQDPGKVFKGKKMAGHMGT RVTTQNI EVVSTDSRGLILVRGAVPGSKGAWILVRDAVKASLPENAPKPA
GL
VEQNNTGKYAPFIRLIVMGISFVATGLTTIFGWEPLPFTDEQMNQGLMLVLSVGLAIYNWYKNNAVTSYGKAKEEAGKEVVGTRQDFKN
RD
VNGKLVTIASLREIGRLLLDIHGQHDNQLLMEDENHLQLLDKFAGAEVESALKTYQEGYQRYVKLLKKLKLQ LSESEQEMAHRLDLIQFQ
LE
AFTGLGAPYWKPDARGTITGLTRGTTIAHIAARALESIAFQSAALL LAMS RDAVASGGAPVSEL RVDGGACVNNLLMQFQADLLGIPVV
RPA
AINCGLVSQSLAACLGFLENGQGPNAACCNVGKTLRNLTPPTQDKRTACRCMKSAASAIPGINHKYSAALPGKCGVSIPGPVGPQADCS
QIH
AITCGQVNSAVGPCLTYARGGAGPSAACCSGVRSLKAAASSTADRRTACNCLKNAARGIKGLNAGNAASIPSKCGVSVPYTISASIDCS
RVS
ATNSSKPQPTYEELEEEEQEDQFDLTVGITDPEKIGDGMNAYVAYKVTTQTS LPLFRSKQFAVKRRFSDFLGLYEKLSEKHSQNGFIVP
PPP
AVPCSTVDMKAAACVGFATGKDSKPSSACCTGLQQLAQTVKSVDDKKAICRCLKASSKSLGIKDQFLSKIP AACNIKVGFVPVSTATNCE
TIH
GNLKNLYMGCYGIGITRIAAA VIEQHHDKNGIWPDSIAPFEVVILPINMKKDNKIKIIAHFLYKKFKKTGIDVILDDRDERPGVMFNE
VDL
LTTQAETVTRSAVLLLTSDRGLAGAFNSNAIKAAEQLTARLEAEGREVD TYIVGRRGA AHYFNFRERKVT ELWTGFTDEPTYADAKKVAG
PLI
MRVLYIIAYDITDARRLGQIRYFLKGYSTGGQKSVYECFLEREELKFIISKIKRLINPNEDRVHIFRIDGRSKVITLGI AVPPIDPEYF
YIG

VAPEEHPVLLTEAPLNPKANREKMTQIMFETFNSPAMYVAIQAVLSLYASGRTTGIVLDSGDGVSHTVPIYEGYALPHAILRLDLAGRD
LTD
VNCGQVNKALSSCVPLTGFDTTPSLTCCAGVMLLKLRLAPTVDKRIACECVKTAARYPNIREDAASSLPYKCGVINVPISKTTNCH
EIN
VSITNKDIKDKRRYIFSQSSKTTTIKRGDKRISSATRICAVCGRPLSKLVLRGTGPTVVVDHISCKISDIVRLNVCEDIRSCYAYSSKK
GES
AHLDEELQATLHDFRHQILQTRGALS LQGSIMTVGEKVFSSNGQSITFDALQIEACARAGGRIAVPRNPEENEAIASFVKKYNTYAYVGL
TEGP
AISC GAVTSDLSPCLTYLTGGPGPSPQCCGGVKLLAAANTTPDRQAACNCLKSAAGSITKLNTNNAALPGKCGVDIPYKISTSTNCN
TVKF
AISC GQVASAIAPCISYARGQGSAPSAGCCSGVRS LNNAARTTADRRAACNCLKNAAAGVSGLNAGNAASIPSKCGVSIPYTISTSTDC
SRVN
ALNCGQVDSKMKPCLTYVQGGPGGPGSLCCNGVRDLHNQAQSSGDRQTVCNCLKGIARGIHN LNLNNAASIPSKCNVNPYTISPDI DC
SRIY
ELQNSKEIELTG FQGLSSCSEETTAPSSSYSSKQASVFIENG DNETSTYRPQNVLTNLNSLYTT FEDARAQKG MVRHKS EDLQSF L
EKYP
MAKFFNYTIVQGLLMSVLLASCVIHAHII SGETEEVSNIGSPTVMVTMGANRKII GDNKNLLCYLKALEYCCERTKQCYDDIKKCLE
HCHG
MKTLLVLLSALVLLALQVQAEPTPKTDEGKTDEQPGKEDQVVSIEGQGDPAFQDGVLRDLKCFRAKSCNWGEGIMGICNKRYGMLI
LCCR
PRRGLPALPTQPGLLPRDARAPLPRQGSVRFD RRRGLLYLRPPPPSPGPRKSRTVRAPPPPPPLRLGRPLSRRLSLSSPAWAFNAVAL
AVLV
QNICPRVNRIVTPCVAYGLGRAPIAPCCRALNDLRFVNTRNLRRACRCLVGVVNRNPGLRRNPRFQNI PRDCRNTFVRPFWWRPRIQC
GRIN
RANAISPEVIARLSERYPNFIGFKDGTG NIEHLAKITTL CGDRLFYLGGLPTAETFALPLLQMG MSTYSSAMFNFIPDFALS FYADVR
AQDS
TLEGELAKAMFAIPAVKGVEFGAGFEASRLRGSKNNDPFSIQANHIRTTSNKCGGV LGGISDGMPLQFRVAVKPTPSISLSQPTVNIDT
MTNT
AGLKNLDVLELFSQECLNKPINHQS NLVVPNTVKNTDGSLVTP EII SDDGDY EKP NVKWHLPEFTNEVS FIFYQPTIGKAKARFHGRV
TQPLK
EVTQG PLCPTGAFSALHLKGT DCTNPSSSLRP SSESTRNQIVGRNSPSIQQEDLDMLDNHEAEDKDH MHPSTSGMTDSSSGSASSHPTF
KQNT
FLADWYVNNDEYLKPELDEKWQ QIIKL RKEVNKKLEKARQGENKIIGNSLDAKVS LYTEDNTLKEFIKENLELLKIVFIVSDLEVVD SA
DGNYT
GEVWVMGKSGKERRLP IGRNAVAWIEHWLDLRDLFG SEDDALFLSKLGKRISARNVQKRFAEWGIKQGLNNHVHPHLRHSFATHMLES
SGDLR
IEDLQSGARIQPGE EKQDPLDFALWKA AKPGEPSWESPWGQGRPGWHLECSAMSEKYFELPFDI HGGGQDLSFPHHENEIAQSEAATGK
EMARF
KPIVYGNVARYFGKKREEDGH THQWTVYVKPYRNEDMSAYVKKIQFKLHESYGNPLRVVTKPPYEITETGWGEFEIIIKIFFIDPNERP
VTLYH
MELVVWKDGDNWRVFADDAVPALDGTVD RVAEGLATLYRKEKEPVVFTRRKTA FYDGQYWL MWKYPTYGRAYGKEASDEMIKEPSAIA
LLAGY
MRELYLVIA YDTPDDRRLARLAKLLKGFGERRQYSVF EARLTREQWAHLKGKLEALVNKEEDVLAVYFLPPEAVGRTWRIGHEGLKRL E
DPDFV
NPTGIISDADKIPFHPYYTVKDALGF LLLLIALLLTLVLFSPDLLGDPDNYTPANPLNTPPHIKPEWYFLFAYAILRSIPNKLGGVLALV
LSIAI
QLES LPPGSIVGTSSLRRLAQLRYKPHLDFKDIRGNVITRIEKLDSGEFDCIILAAAGLKR LGFESRVHQIIPNEISLHAVGQ GALGI
ECKSD
RGVDVKFIFSGGNAFDVLAKGGGKGQALAYLLKKLKT EGKLPI NTLACGDSGNDTELFTIPNVYGMVSN AQEELLEWYAENAKDNANI
IHASE
SYNSTAALVSTILGTPIKMYNESGQPLFDVGNMNPKNII EKMI EIEIPIFKSDYRN NESPD FDSWNERSNLKIVSVNDCHAICIFKFE
NNWWC
VTRYKVEWSMEDFSPLAGEIIMDNLQTLRCTITGLTMGQQYFVQVSAYNMKGWGAQTTPACASPSNWKDYDDREPRHKQSEVLEG
LLQQV
WQVPTGRRDGRVSLASNVNLPSPSDSLAIQQRKFSAFRLNTRDLVTLVGGGHTIGTAACGFITNRIFNSSGNTADPTMDQTFVPQLQR
LCPQN
YPEPGITFRDLTPVFADGAALRAVVDALVEPFAGQF DAVAGVEARGFLLAAAAAYATGTGVITVRKAGKLPRAVYTEHYSLEYGTAAL E
LHRDD
AALAEFGIRGIDLTRIESRPTRELGTYLFFVDCVGHIDDEAVAEALKAVHRRCADVRYLGSWPTGPAAGAPPLVDEASRWLARLRAG
KPEQTL
GDAGVQACFGRSREYQLNDSAKYYFDSIQRMAEPSYLP TDQDVLRSRVKTTGITETHFKIGELNYKLF DVGGRSERKKWIHCFENVTA
IIFLVA
HATGRALGVRANDLNRQMAKNDKDNELFAKEFKKVLQNSKDGDRQNQALKAFMSNPVQKKNLE NVLANNGNTEVISPTL FVLLMYLLSM
ITAYIF
LILDFGSQYTQLVARRVRELGVYCELWAWDVTEAQIRDFNPSGIILSGGPESTTEENSPRAPQYVFEAGVPVFGVCYGMQTMAMQLGGH
VEASNE
LRVWIEDKVGLLTGYSTEPLEGYKCVNIDPSESLEMLGGMLDFHNYYYDGEKVYRD TNND FQKFL EEEANKPPEPSKEEMEERLAKLEA
LLADLL
MAEIIKFVYIMILCVSLLLIEVAGEECVTDADCDKLYPDIRKPLMCSIGECYSLYKGKFSLSII SKTSFSLMVYNVVTLVICLRLAYIS

LLLKFL
MILVTYDVNTVEPGRRRLRQVAKACQDYGQRVQNSVFEVEVDPARWVALKARLEAIIIDPALDSLRYD LGANWQRRVDHVGAKPAVDL
HGPLIL
MKLQATARVAGLLFLVLLLALPSLRVSMAGSGFCDGKCAVRCSKASRHHDC LKYCGICCATCHNCVPSGTAGNKDEPCYRDMTTGHGNR
TRPKCP
MPYVVVFYDVSDNKRRLAKTLQSLGLVRVQSVFMGRGGYTKAKEAIRAASRIVDARTDSVVALVPEDYARRMLVYGGIMSDPKQK
QAVRVV
MRGSLDYNNYLYQTIENRPTEELDVLYDGLYKKAGDLFAIDNFQGVKEGRLILKILKAIREEINSRIDEEIDLYLYNIYDSISDEDKRV
NWLIEV
VKVVDIVDTFRLQEQQPFDDKKQFLGFIKRYIKNLATKLSEERQAEFKKNVEGAAMLVSKLSDLQFFVGESMHDDGSMVFAYYKDGATD
PTFLYF
YEALVTSILGKLTGLWHNDSVDFMGHICYFRRRPKIRRFKLYHEGKFWCPGWAPFEGRSRTKSRSGSSREATKDFVRKALQNGLVTQQD
ASLWLN
IDTMGAYPGIGAEERGQSEAIGAALAAMARVGVPIVATVIGEGSGGALALGVANRVLVLEFSCYSVISPEGCAAILWKDGSRADEAAA
RLKITAP
IGTFSEGKIEEGLKRYALISALQDSRFTPIEREELSQRRCGNLLSQFKTIYSSEGTGNSGDIWNWEIGKHGIEIKFRHPKTNSRMSAT
FLPEVIP
KNRRREFGLLERPVLPPSVVIDTASYKIFVSGKSGVGKTALVAKLAGLEVPIVHHETTGIQTTVVFWPAKLKASDCVVMFRFEFWDGCE
SALKKFD
LNHPVKNLATGKVAITSQNHGYAVEEESVENTDLEITHVALNDGTVEGLRHKKFPAFTVQYHPEASAGPEDANDLFEDFLTMENFKKE
GEELCQN
MAISKSTVVVVILCFILIQELGIYGEDPHMDAAKKIDCGGKCNRSRCSKARRQKMCIRACNSCKKRCVPPGTSGNRDLCPYARLTTH
GGKLKCP
MALKKSNARTNTAAQSDDTRAASF INISIGTRGGDPVRLNGIPLRLSEAVEAQLHEYLAELAKDDKDLAKRIENLRSRLILSFRVVRDK
SELQLDL
MKATILLAVLVAVFVAGTEAHSHACTSYWCGKFCGTASCTHYLCRVLHPGKMCACVHCSRNNPFRVNQVAKSINDLDYTPIMKSMENL
DNGMDML
MKATMLLAVVVAVFVAGTEAHPHVCTSYYSKFCGTAGCTRYGCRNLHRGKLCFCLHCSRVKFPFGATQDAKSMNELEYTPIMKSMENL
DNGMDML
MLEEDLKKREAI VSKKEALLQEKS LLENKKLRSSQALSTDGLKISARLNLDDQELSEKSLQLESSPTEEKMKISEQVQALQRERDQLQR
QRNSVDE
MQIGKNMVETPKLDYV IIFFFLYFFFRQMIILRLNTTFRPLNFKMLRFWQNRNIMKHRGQKVHFSLILSDCKTNKDCPKLRRANVRCR
KSYCVPI
MRAIVFVLIFAIAFAATREGSILCNLCKDTVNL IENLLTVDGAQAVRQYIDNLCAKADGFLGTL CNKILSFGVDELVKLIENHVDPVVI
CEKIHAC
MRTLTLTLLTLLALHTQAESPQGSSTEEALDQEQLVMEDQDISISFGDGKTALQDADV KSGVTCYCRLLSCQFGERLAGSCRSGGV
TYPLCCH
MYILITYDVSTETEAGKKRLRKVAQVCKDFGQRVQKSVFECSVNEAQFEQLKHRLLCIDEKSDSLRIYRLREPAKKYIQEYGVNLTID
FDAPLVL
TASRDLRFLQEVLEKQMTARGYMVGPNGPVNLQIIVSQLYADVSQGNVRYNIATKADIAIIATAQNGNKMTKNYRASYNVEGAFQASN
KNIADAV
VPTVICFLAMCLVAITGAYPQEPVLADEAQSVANS LFDELPEESYQAAVENLRLKRATCDLLSGFGVGDSACAAHCIARGNRGGYCNSK
KVCVCPI
YAGCVS QLFYHFLGGTECFLYTMAYDRFVAICYPLRYSVIMSHRICAFLAMGTAVFGCIHSTFLTTLTFQLPYCGPKDVNYYFCDIP
VVMKLAC
LREMAQKARERRAGIKTHVEKEDGEARERDEIRHRRKERQHDRNLSRAAPDKRSKLQRNENRDI SEVIALGVPNPRTSNEVQYDQRLF
NQSKGMS
MAARSYSPIMVALS LLLLVTFSNVAEAYTRSGTLRPSDCKPKCTYRCSATSHKKPCMFFCQKCCAKCLCVPPGT YGNKQICPCYN SWKT
KEGGPKCP
MAEPRRWYLITYDIRDPKRWRKVHALLKGYGEWLQLSVFRCSLTDRDREKLRLWELSRMDAVDTLLVIGLCGGCVERVRAINAKEDWPE
EPAPFKVL
MELKSGLSILLCFGICIAVINAGCFEDWSRCS PSTSRGTGVLWRDCDSYCKVCFKADRGECDSPSLNCPQRLPNNKQCRCINARTAKD
NRNPTCWA
MEVVEEGIMKSTGERITRDN IKEGMKVVCVKS IYESQGYFTVGKEYEVMGNCGYLG IKNDRDGF IWDCAIFNEDKFVFEILEEPSEK
SFFKSF RK
QQA EIAMDEADV IIFMVNGREGVTSAD EEVAKILYRTKKPVVLAVNKL DNT EMRANIYDFYALGFGEPPYISGTHGLGLD LLDACA EH
FKNIPETK
RFKGGDIVDTVTL SRRGVDFS YVDGNEYV FMDKEDYTPYTF TKDQIEEELLFMPEGGM PDMQVLTWDGQLLAL ELPQTVDLEIVETAPG
IKGASASA
DDKLDVILDRKEATLSFINGDEYTFMDTTDYTMELNAEDIEAVLPYIEEGMEDVCEAVFFEGRLVSVELPTTISRKV VYTENAARGD
TSGKVMKPA
DGSRGSESSPGMVSV DPLPKAEAPAFLSRAGSKGMFGAHPGHSYGDPPGPPPAQLFQESEL FYLAQESQVPSRTRAPRLPEPGSSRAG
DSSEGYEEE
GWL DIVKAI VVPAARETIKTQEITLLDHYCTLSRSPYIKSLELHYRAEVTCPGWTIIRGRGSNHRNP TNSGKDALKDFMTQAVAAGLVT
KEEAAPWLN
IDGQWHVHQLK KAVLLSFRIFDNAVIDGAETKYFDKVLKFAEYDEARFKA EAIRVVPSATVRKGSFIGKNTVLMPSYVNLGAYVDEG
TMVDTWATV
ITLQPVVRACVFLATAVA AVIMGLNKQSYTTVVAIVGTRPVTQTFTAKFKDTPAFVFFVIANAIASGYNLMVLVTRRILQRRASLSVH
LLDMVILT L

MAISKALIASLLISLLVLQLVQADVESSQKKNGYAKKIDCGSACVARCRLSRRPRLCHRACGTCCYRCNCVPPGTYGNYDKCQCYASLT
THGGRRKCP
MALKVGGLTLEKTKRKKSDKYAKEEETARRNRENIKKWFGNAWDGVKSKTGEAFSKMGRNANHFGGEMKKNVERNQRDSKQIKFRLEL
SQKFCRIPH
MSNLKWFSGGDDRRKKAIEVITELLDDLEIDLGNESLRKVLGSYLEKLKNEGTSVPLVLSRMNIEISNAIKKDGVSLENQSKKLKELI
SISNIRYGY
MSTNLNLTGRLFDIKIIPITNGKLTGFRVRMTLFNMDCPNTVLTHVVMCGDEYTAKS LKTAILQRRRLNLKHHWTEVNDLPGACTNRL
KTKPFTLEI
MTKTELQYKKAIGVAIFATSEEDKKQLGNVAPFSIYEILEIDLNKNRVYYALNCGERHAVCFTKLRKEEETGDNFILINKQPFFLKDIH
KGLTWSKSL
TPAQSQRPRLIDHTGDDDAGPLGTNALSTLPPLEINSDEAMQLGYMPNRDSFEREYDPTAEQLISNISLSSSEDTEVDVMLKLAHVDIYT
RRLRERARR
TTSEAAISCGQVSSAIALCLSYARGQGFAPSAGCCSGVRSLSAARTTADRRACNCLKNAARGISGLNAGNAASIPSKCGVSVPYTIS
TSTDCSRVS
AEDAAEERILDALLPPAKNQWGEVESHSSTRQAFRKKLREGQLDDKEIEIDVSAGVSMGVEIMAPPGMEEMTNQLQSLFQNLGSDK
TKKRKMKIKD
FSFEKAESLYTRAPHLFTWLKGRFVTNSISYWTFLSGGQNKYSYTNSSINEGSFRGQD TDYGGTSSTINIPSNSYVYNLWTENYEYI
YPWGD PVNIT
GYREPIFRKVLTTAALDSGINYT LISGPSLDPSKFKEIPKNVQILRFVGDTPYIRSSDAVIAPGGHSTMMEALSFGTPILSFPDEGHS
EQENNAAVIE
IIHLSQANDYCKIKCSSGVHTVCQYGESTKPSKNCAGKLIKSVGPTEEKKLIVEEHNFRQKVAKGLETRGNPGPQPAASNMMNLVWN
DELA KIAQVW
MAIFRSTLVLLLILFCLTTFELHVHAAEDSQVGEVVKIDCGGRCKGRCSKSSRPNLCLRACNSCCYRCNCVPPGTAGNHHLCPCYASI
TTRGGRLKCP
MALSKLIASLLASLLLLHFVDADQSAHAQTQGSLLQQIDCNGACAARCRLSSRPRLCQRACGTCCRRNCVPPGTAGNQEVCPCYASL
TTHGGKRKCP
MGSKVYYVLMALSFYVVLCAFPNRNSVAFEKQHESLPHADYIVTQKTESHEKTTAEPKLPGRICWQYEEVTEDAICQEHCI PKGYSYG
LCISNTCSCI
MMMDKQVEEVKKHYPIVEDWSVIVARKEDDCMTVTDAVPFILAGYKNVSYEMDDIVVLCSEPIGLTWEDVRFLKNHEGSVSFEEIGYED
KAMVYHVDLG
MQLIKEYEHAQPLKFKGWQVSEDHEQNVQALQEMLGSPFFITAAFGDNADGERVYSLSNFNEGATLDVKAGEWIFVNEEGSIEGWSAEV
GAVRFKEIQA
QASAIGQFKDRIIGATNPGNASITILHMQPADSGIYICDVNNPPDFFGKNQGILDVTVLVKPSKPFCSIQGRPEAGHPISLSCLSAFGT
PSPVYYWYKI
REIAVDLRDDWPQALRDSGFDAEAPSAWIAEGLLIYLPAAAQEQLFTGIDSLACLGSHVAVEDGAPLPDAEFAAKLEERAAVAAGGER
PFFQLLYNER
VKVTLRGEDAHAFDLTALELADISKSQFDDTGNL SFGVEDHTDFPSQEYDPNIGIYGLDVTTTIVRPGYRVSKRDKATASIPSKHRMTA
EDATAFLETN
MAVAKPPLQTA AVL LLLLLLVVAAASWLQTVDAASGFCSSKCSVRCGRAASARARGACMRSCGLCCEECNCVPTRPDRVNECPCYRDML
TAGPRKRPKCP
MMTKCQKRASIQGLWLLSMVLLASSSLVCASMAVDGQTKEDINATSVTSMNMTRSSSASYNMTGGGGELNRGPCVVRSGFYWCQNIGYP
TMSECLKNCES